A3

REMODEL

GENERAL NOTES:

1. GENERAL NOTES APPLY TO ALL DRAWINGS
2. DO NOT SCALE DRAWINGS. ANY DIMENSIONAL INFORMATION REQUIRED WHICH IS NOT INDICATED ON DRAWING DIMENSION STRINGS SHALL BE OBTAINED FROM THE ARCHITECT.
3. DIMENSIONS SHOWN ARE FINISH SURFACE OF EXISTING CONSTRUCTION, UNLESS NOTED OTHERWISE .
4. CONTRACTOR SHALL VERIFY ALL EXISTING DIMENSIONS AND CONDITIONS (I.E EXISTING MATERIALS, FRAMING MEMBER SIZES AND LOCATIONS, METHODS OF CONSTRUCTION). IF DISCREPANCIES ARE FOUND, NOTIFY ARCHITECT BEFORE PROCEEDING WITH WORK.
5. CONTRACTOR SHALL MAINTAIN THE IMMEDIATE CONSTRUCTION SITE IN A SECURE, CLEAN AND SAFE MANNER.
6. PROTECTION : CONTRACTORS SHALL BE SOLELY RESPONSIBLE FOR TAKING ALL STEPS NECESSARY TO PROTECT THE PUBLIC FROM INJURY AND ADJACENT PROPERTY DAMAGES DURING CONSTRUCTION AS REQUIRED BY LOCAL CODES.

SCOPE OF WORK:

1. PLUMBING
2. MECHANICAL
3. ELECTRICAL

APPLICABLE CODES:

2021 IBC – International Building Code
2009 ANSI A117.1 – Accessibility Code
2018 IECC – International Energy Conservation Code
2018 IRC – International Residential Code
2018 UPC – Uniform Plumbing Code
2018 UMC – Uniform Mechanical Code
2017 NEC – National Electrical Code

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OWNER SIGNATURE

REVISION STATUS

Rev	DATE	DESCRIPTION	DRAWN BY
00	12-10-2023	ISSUED FOR APPROVAL	A.S
01			
02			
03			
04			

CLIENT/CONSULTANT APPROVAL STATUS

APPROVED	**(A)**
APPROVED WITH NOTES	**(B)**
RESUBMIT	**(C)**
DISAPPROVED	**(D)**

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DRAWING STATUS:

DISCIPLINE:
ELECTRICAL

PROJECT TITLE:
23-5550F_NCARROLLTON636

DRAWING TITLE:
COVER SHEET

DRAWN BY:	CHECKED BY:	DATE:
A.S	N.B	12 OCT. 2023
JOB NO	SHT NO	SCALE:
632-021	EL-00	NTS @A3

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BASIC ELECTRICAL REQUIREMENTS:

1. GENERAL: ALL ELECTRICAL WORK SHALL CONFIRM TO THE REQUIREMENTS OF THE NEC NATIONAL ELECTRICAL CODE (NEC) AND NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA) STANDARDS UNLESS MORE STRINGENT REQUIREMENTS ARE INDICATED. ALL ELECTRICAL WORK SHALL COMPLY WITH ADA RECOMMENDATIONS. ALL ELECTRICAL WORK SHALL BE PERFORMED IN A NEAT AND WORKMAN LINK MANNER BY A LICENSED ELECTRICIAN, OR A CERTIFIED APPRENTICE WORKING UNDER THE DIRECT SUPERVISION OF A LICENSED ELECTRICIAN, USING THE BEST METHODS KNOWN TO THE TRADE AND SHALL PRESENT A NEAT AND PROFESSIONAL APPEARANCE WHEN COMPLETED. THE OWNER RESERVES THE RIGHT TO CHANGE, WITHOUT ADDITIONAL COST, THE LOCATION OF ANY APPARATUS OR OUTLET, PROVIDED SUCH CHANGE DOES NOT ADD MORE THAN 10 FT TO THE FEEDER ADN IS ORDERED BEFORE INSTALLATION OF THE AFFECTED PORTION OF THE WORK IS COMMENCED. CONTRACTOR SHALL BRING ALL CONFLICTS ON THE DRAWINGS TO THE OWNERS ATTENTION FOR HIS RESOLUTION PRIOR TO PERFORMING THAT WORK.

2. THE CONTRACTOR SHALL SURVEY THE PROJECT SITE PRIOR TO THE BID TO ASSESS ACTUAL FIELD CONDITIONS. FAILURE TO PERFORM THIS INSPECTION BINDS THE CONTRACTOR TO PERFORM THE WORK WITHOUT EXTRA CHARGES DESPITE THE IGNORANCE OF REASONABLY ANTICIPATED CONDITIONS.

3. ROUGH–INS: THE CONTRACTOR SHALL VERIFY AND COORDINATE THE ROUGH–IN REQUIREMENTS OF EACH ITEM OF EQUIPMENT WITH THE CONTRACTOR SUPPLYING THE EQUIPMENT.

4. INSTALLATION: THE ELECTRICAL DRAWINGS INDICATE THE EXTENT AND GENERAL LOCATION AND ARRANGEMENT OF EQUIPMENT AND MATERIALS. THE CONTRACTOR SHALL BECOME FAMILIAR WITH ALL DETAILS OF THE WORK AND VERIFY ALL DIMENSIONS IN THE FIELD SO THAT EQUIPMENTS AND MATERIALS WILL BE PROPERLY LOCATED AND READILY ACCESSIBLE. THE CONTRACTOR SHALL SEQUENCE, COORDINATE, AND INTEGRATE THE VARIOUS ELEMENTS OF ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS. COMPLY WITH THE FOLLOWING REQUIREMENTS:

   A. VERIFY ALL DIMENSIONS BY FIELD MEASUREMENTS.

   B. COORDINATE ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS INSTALLATION WITH OTHER BUILDING COMPONENTS AND TRADES.

   C. ARRANGE FOR CHASES, SLOTS, AND OPENINGS IN OTHER BUILDING COMPONENTS DURING PROGRESS OF CONSTRUCTION, TO ALLOW FOR INSTALLATION OF ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS.

   D. SEQUENCE, COORDINATE, AND INTEGRATE INSTALLATION OF ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS FOR EFFICIENT FLOW OF THE WORK.

   E. WHERE MOUNTING HEIGHTS ARE NOT INDICATED, INSTALL ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS TO PROVIDE MAXIMUM HEADROOM POSSIBLE.

   F. INSTALL ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS TO CONFORM WITH APPROVED SUBMITTAL DATA TO THE GREATEST EXTENT POSSIBLE. CONFORM TO THE ARRANGEMENTS INDICATED ON THE ELECTRICAL DRAWINGS, RECOGNIZING THAT PORTIONS OF THE WORK ARE SHOWN ONLY IN DIAGRAMMATIC FORM. WHERE COORDINATION REQUIREMENTS CONFLICT WITH INDIVIDUAL SYSTEM REQUIREMENTS, REFER CONFLICT TO THE ARCHITECT/ENGINEER OR OWNER'S REPRESENTATIVE FOR RESOLUTION.

   G. IN GENERAL, INSTALL ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS LEVEL AND PLUMB, PARALLEL AND PERPENDICULAR TO BUILDING LINES AND/OR FEATURES AND/OR OTHER BUILDING SYSTEMS.

   H. INSTALL ELECTRICAL EQUIPMENT TO FACILITATE SERVICING, MAINTENANCE, AND REPAIR OR REPLACEMENT OF EQUIPMENT AND COMPONENT PARTS. AS MUCH AS PRACTICAL, CONNECT EQUIPMENT FOR EASE OF DISCONNECTING, WITH MINIMUM OF INTERFERENCE WITH OTHER INSTALLATIONS.

   I. INSTALL ELECTRICAL SYSTEMS, EQUIPMENT, AND MATERIALS GIVING RIGHT–OF–WAY PRIORITY TO SYSTEMS REQUIRED TO BE INSTALLED AT A SPECIFIC SLOPE (INCL. SPRINKLER SYSTEMS).

5. CUTTING AND PATCHING: ALL ELECTRICAL WORK SHALL BE CAREFULLY LAID OUT IN ADVANCE, AND WHERE CUTTING, CHANNELING, CHASING, OR DRILLING OF FLOORS, WALLS, PARTITIONS, CEILINGS, OR OTHER SURFACES IS NECESSARY FOR THE PROPER INSTALLATION, SUPPORT, OR ANCHORAGE OF CONDUIT OR OTHER ELECTRICAL WORK, THIS WORK SHALL BE CAREFULLY DONE. ANY RESULTING DAMAGE TO THE BUILDING OR OTHER SYSTEMS, EQUIPMENT, OR MATERIALS SHALL BE REQUIRED BY SKILLED MECHANICS OF THE TRADES INVOLVED, AT NO ADDITIONAL COST TO THE OWNER.

6. PRODUCTS: SYSTEMS, EQUIPMENT, AND MATERIALS DESCRIBED ON THE ELECTRICAL DRAWINGS ESTABLISH THE MINIMUM STANDARDS FOR QUALITY AND STYLE AND SHALL BE THE BASIS OF THE BID. ALL SYSTEMS, EQUIPMENT, AND MATERIALS SHALL BE NEW AND SHALL BEAR THE UL LABEL OR BE UL LISTED, WHERE APPLICABLE, AND SHALL BE INSTALLED IN ACCORDANCE WITH THE NEC AND NEMA STANDARDS.

7. SUBSTITUTIONS: WHERE SYSTEMS, EQUIPMENT, OR MATERIALS ARE SPECIFIED BY MANUFACTURER OR BRAND NAME AND CATALOG NUMBER, SUCH SPECIFICATION SHALL ESTABLISH THE MINIMUM STANDARDS FOR QUALITY AND STYLE AND SHALL BE THE BASIS OF THE BID. SYSTEMS, EQUIPMENT, AND MATERIALS SO SPECIFIED SHALL BE FURNISHED UNDER THE CONTRACT UNLESS CHANGED BY WRITTEN AGREEMENT. SHOULD THE CONTRACTOR PROPOSE TO FURNISH PRODUCTS OTHER THAN THOSE SPECIFIED, AS PERMITTED BY "OR APPROVED EQUAL" CLAUSES, HE SHALL SUBMIT A WRITTEN REQUEST FOR SAID SUBSTITUTIONS THROUGH APPROPRIATE CHANNELS TO THE ARCHITECT ENGINEER FOR HIS REVIEW. SUCH REQUEST SHALL BE ACCOMPANIED WITH COMPLETE DESCRIPTIVE LITERATURE INCLUDING, BUT NOT LIMITED TO, CATALOG CUT SHEETS, BROCHURES, CIRCULARS, SPECIFICATIONS, PERFORMANCE DATA, INSTALLATION INSTRUCTIONS, SHOP DRAWINGS, AND OTHER PRINTED INFORMATION IN SUFFICIENT DETAIL AND SCOPE TO VERIFY COMPLIANCE WITH THE REQUIREMENTS OF THE CONTRACT. DESCRIPTIVE LITERATURE ON PROPOSED SUBSTITUTIONS SHALL BE CLEAR, CONCISE, AND LOGICALLY ARRANGED. ALL DATA WHICH IS. AND IS NOT, APPLICABLE SHALL BE CLEARLY IDENTIFIED AS SUCH. IF REQUESTED, THE CONTRACTOR SHALL SUBMIT SAMPLES OF BOTH SPECIFIED AND PROPOSED ITEMS FOR INSPECTION. DESCRIPTIVE LITERATURE ON PROPOSED SUBSTITUTIONS SHALL BE RETURNED WITHOUT REVIEW IF NOT PROPERLY PREPARED. ACCEPTANCE OR REJECTION OF PROPOSED SUBSTITUTIONS SHALL BE UP TO THE DESCRIPTION OF THE ARCHITECT/ENGINEER AND/OR THE OWNER.

8. SUBMITTALS: THE CONTRACTOR SHALL FOLLOW THE GENERAL PROVISIONS OF THE CONTRACT AND ESTABLISHED PROCEDURES. SUBMITTALS SHALL CONSIST OF COMPLETE DESCRIPTIVE LITERATURE INCLUDING, BUT NOT LIMITED TO, CATALOG CUT SHEETS, BROCHURES, CIRCULARS, SPECIFICATIONS, PERFORMANCE DATA, INSTALLATION INSTRUCTIONS, SHOP DRAWINGS, AND OTHER PRINTED INFORMATION IN SUFFICIENT DETAIL AND SCOPE TO VERIFY COMPLIANCE WITH THE REQUIREMENTS OF THE CONTRACT. DESCRIPTIVE LITERATURE SHALL BE CLEAR, CONCISE, AND LOGICALLY ARRANGED. ALL DATA WHICH IS, AND IS NOT, APPLICABLE SHALL BE CLEARLY IDENTIFIED AS SUCH. IF REQUESTED, THE CONTRACTOR SHALL SUBMIT SAMPLES OF SPECIFIED ITEMS FOR INSPECTION. DESCRIPTIVE LITERATURE SHALL BE RETURNED WITHOUT REVIEW IF NOT PROPERLY PREPARED. THE FOLLOWING SYSTEMS, EQUIPMENT, AND MATERIALS, AS A MINIMUM, REQUIRE SUBMITTALS:

   A. ANY PROPOSED SUBSTITUTIONS:

      B. WIRING DEVICES.

      C. PANELBOARDS.

      D. DISCONNECT SWITCHES.

      E. CIRCUIT BREAKERS.

      F. LIGHTING FIXTURES INCLUDING BALLASTS.

9. RECORD DRAWINGS: THE CONTRACTOR SHALL MAINTAIN AT THE SITE A CLEAN, UNDAMAGED SET OF BLUE– OR BLACK–LINE WHITE PRINTS OF CONTRACT DRAWINGS. THIS RECORD SET OF CONTRACT DRAWINGS SHALL BE MARKED TO SHOW THE ACTUAL INSTALLATION AND WHERE THE ACTUAL INSTALLATION VARIES SUBSTANTIALLY FROM THE ELECTRICAL WORK AS ORIGINALLY SHOWN. MARK WHICHEVER DRAWINGS ARE MOST CAPABLE OF SHOWING CONDITIONS FULLY AND ACCURATELY. GIVE PARTICULAR ATTENTION TO CONCEALED ELEMENTS THAT WOULD BE DIFFICULT TO MEASURE AND RECORD AT A LATER DATE. MARK RECORD DRAWINGS WITH A RED ERASABLE PENCIL; USE OTHER COLORS TO DISTINGUISH BETWEEN VARIATIONS IN SEPARATE CATEGORIES OF THE ELECTRICAL WORK. NOTE CONTRACT MODIFICATIONS AND APPROVED SUBSTITUTIONS WHERE APPLICABLE.

10. PROTECTION OF INSTALLED SYSTEMS, EQUIPMENT, AND MATERIALS: PROTECT INSTALLED SYSTEMS, EQUIPMENT, AND MATERIALS FROM DAMAGE UNTIL FINAL ACCEPTANCE BY THE OWNER. REPAIR OR REPLACE, AT NO ADDITIONAL COST TO THE OWNER, DAMAGED SYSTEMS, EQUIPMENT, AND MATERIALS TO THE SATISFACTION OF THE ARCHITECT/ ENGINEER AND/ OR OWNER.

11. CLEANING: UPON COMPLETION OF INSTALLATION, INSPECT INTERIOR AND EXTERIOR OF ALL ELECTRICAL EQUIPMENT. REMOVE PAINT SPLATTERS AND OTHER SPOTS, DIRT, AND DEBRIS. TOUCH–UP SCRATCHES AND MARKS OF FINISH TO MATCH ORIGINAL FINISH.

12. CERTIFICATIONS: THE FOLLOWING SHALL BE OBTAINED AND SUBMITTED TO THE OWNER PRIOR TO FINAL PAYMENT:

    A. ELECTRICAL SYSTEMS: A CERTIFICATE OF FINAL INSPECTION AND APPROVAL BY THE AUTHORITIES HAVING JURISDICTION

13. GUARANTEE: THE CONTRACTOR SHALL SUBMIT A WRITTEN GUARANTEE TO THE OWNER, PRIOR TO FINAL PAYMENT, THAT WARRANTS THE INSTALLATION SHALL REMAIN FREE OF DEFECTS FOR A PERIOD OF ONE YEAR AFTER FINAL ACCEPTANCE BY THE OWNER. THE GUARANTEE SHALL STATE THAT THE OWNER IS NOT LIABLE FOR PARTS AND LABOR COSTS INCURRED BY THE CONTRACTOR IN THE REPAIR OF ACTUAL PRODUCT OR INSTALLATION DEFECTS. THE GUARANTEE SHALL STATE THAT THE ON–SITE RESPONSE TIME TO REQUESTS FOR ASSISTANCE WILL BE 24 HOURS FOR NON–EMERGENCY CONDITIONS AND 2 HOURS FOR EMERGENCY CONDITIONS.

ELECTRICAL MATERIALS:

1. RACEWAYS:
   - A. RIGID GALVANIZED STEEL (RGS) CONDUIT: ANSI C80.1.
   - B. ELECTRICAL METALLIC TUBING (EMT) AND FITTINGS: ANSI C80.3 WITH COMPRESSION TYPE FITTINGS.
   - C. LIQUIDTIGHT FLEXIBLE METAL CONDUIT: UL 360. FLEXIBLE STEEL CONDUIT WITH PVC JACKET.
   - D. NONMETALLIC CONDUIT AND TUBING (ENT): NEMA TC 13
2. BOXES:
   - A. SHEET METAL: NEMA OS 1.
   - B. CAST METAL: NEMA FB 1, TYPE FD, CAST FRAILLY BOX WITH CASKETED COVER.
   - C. HINGED COVER ENCLOSURES: NEMA 250, GALVANIZED STEEL ENCLOSURE WITH CONTINUOUS HINGE COVER, QUICK RELEASE TYPE LATCHES, REMOVABLE INTERIOR PANEL, AND MANUFACTURER'S STANDARD GRAY ENAMEL INSIDE AND OUT.
3. WIRES AND CABLES:
   - A. CONDUCTOR MATERIAL: ANNEALED COPPER.
   - B. INSULATION: THHN/THWN CONFORMING TO WC 5.
   - C. CONDUCTORS #16 AWG AND SMALLER SHALL BE SOLID; CONDUCTORS #14 AWG AND LARGER SHALL BE STRANDED. MC CABLE IS PERMITTED FOR BRANCH CIRCUITRY, AND SHALL UTILIZE SOLID CONDUCTORS.
   - D. GROUND CONDUCTORS #10 AWG AND SMALLER SHALL HAVE GREEN THHN/THWN INSULATION.
4. WIRING DEVICES:
   - A. GENERAL: COMPLY WITH NEMA WD 1, " GENERAL PURPOSE WIRING DEVICES."
   - B. COLOR: WHITE, BLACK, GRAY, IVORY, OR BROWN TO BE AS SELECTED BY THE ARCHITECT/ENGINEER AND/OR THE OWNER.
   - C. RECEPTACLES: COMPLY WITH UL 498, "ELECTRICAL ATTACHMENT PLUGS AND RECEPTACLES, "HEAVY DUTY SPECIFICATION GRADE EXCEPT AS OTHERWISE INDICATED. GROUND FAULT CIRCUIT INTERRUPTER (GFCI) TYPE RECEPTACLES SHALL COMPLY WITH UL 943, "GROUND FAULT CIRCUIT INTERRUPTERS," WITH INTEGRAL NEMA 5–20R DUPLEX RECEPTACLE DESIGNED FOR INSTALLATION IN A 2–3/4" DEEQ DEVICE BOX WITHOUT ADAPTER.
   - D. TOGGLE SWITCHES: 20A, 120–277V AC, QUIET TYPE, SPECIFICATION GRADE AND SHALL COMPLY WITH UL 20, "GENERAL USE SNAP SWITCHES." SINGLE–POLE, TWO–POLE, THREE–WAY, AND FOUR–WAY AS INDICATED AND/OR REQUIRED.
   - E. DEVICE PLATES: SINGLE AND COMBINATION TYPES WHICH MATE AND MATCH WITH CORRESPONDING WIRING DEVICES.
   - F. MANUFACTURER: SUBJECT TO COMPLIANCE WITH REQUIREMENTS, PROVIDE PRODUCTS AS MANUFACTURED BY HUBBEL INC. OR APPROVED EQUAL BY LEVITON.
5. GROUNDING:
   - A. GROUNDING AND BONDING PRODUCTS: OF TYPES INDICATED AND/OR OF SIZES AND RATINGS TO COMPLY WITH THE NEC. WHERE TYPES, SIZES, RATINGS, AND QUANTITIES ARE IN EXCESS OF NEC REQUIREMENTS, THE MORE STRINGENT REQUIREMENTS AND THE GREATER SIZE, RATING, AND QUANTITY INDICATIONS SHALL GOVERSMOOTH MATCHING NYLON IN ALL AREAS.
   - B. CONDUCTOR MATERIAL: COPPER.
   - C. WIRE AND CABLE CONDUCTORS: CONFIRM TO NEC TABLE 8, EXCEPT AS OTHERWISE INDICATED, FOR CONDUCTOR PROPERTIES, INCLUDING STRANDING.
   - D. CONNECTOR PRODUCTS: UL LISTED AND LABELED AS GROUNDING CONNECTORS FOR THE MATERIALS USED.
6. PANELBOARDS:
   - A. CIRCUIT BREAKERS: PROVIDE TYPE, RATING, AND FEATURES INDICATED. BOLT–ON EXCEPT WHERE PLUG–IN FOR USE ON EXISTING PANELBOARDS (NOT BEING UPGRADED) TANDEM CIRCUIT BREAKERS SHALL NOT BE USED. MULTIPLE CIRCUIT BREAKERS SHALL HAVE AN INTERNAL COMMON TRIP AND A SINGLE HANDLE..
   - B. ENCLOSURES: NEMA TYPE 1, UNLESS OTHERWISE INDICATED.
   - C. FRONT: SECURED TO BOX WITH CONCEALED TRIM CLAMPS EXCEPT AS INDICATED. FRONT FOR SURFACE MOUNTED PANELBOARDS SHALL BE SAME DIMENSIONS AS BOX.
   - D.. DIRECTORY FRAME: METAL WITH CLEAR PLASTIC COVER MOUNTED ON INSIDE OF PANELBOARD DOOR.
   - E. BUS WORK: HARD DRAWN COPPER OF 98% CONDUCTIVITY.
   - F. MAIN AND NEUTRAL PLUGS: COMPRESSION TYPE.
   - G. EQUIPMENT GROUND BUS: ADEQUATE FOR FEEDER AND BRANCH CIRCUIT EQUIPMENT GROUND CONDUCTORS. BONDED TO BOX.
   - H. PROVISIONS FOR FUTURE DEVICES: EQUIP WITH MOUNTING BREAKETS, BUS CONNECTION, AND NECESSARY APPURTENANCES, FOR THE CIRCUIT BREAKER AMPERE RATING INDICATED FOR FUTURE INSTALLATION OF DEVICES.
   - I. MAIN AND SUFFERED LUGS: PROVIDE WHERE INDICATED.
   - J. NAMEPLATE: CUSTOM ENGRAVED PLASTIC LAMINATE, WHITE LETTERS ON BLACK FIELD, FOR EACH PANELBOARD MOUNTED WITH EPOXY OR INDUSTRIAL CEMENT OR ADHESIVE.
   - K. MANUFACTURER: SUBJECT TO COMPLIANCE WITH REQUIREMENTS, PROVIDE PRODUCTS AS MANUFACTURED BY SQUARE–D, SIEMENS OR CUTLER HAMMER CO.
7. DISCONNECTS AND CIRCUIT BREAKERS:
   - A. FUSIBLE SWITCH, 800A AND SMALLER: NEMA KS 1, TYPE HD, CLIPS TO ACCOMMODATE SPECIFIED FUSES, ENCLOSURE SUITABLE FOR THE ENVIRONMENT WHERE INSTALLED, HANDLE LOCKABLE WITH TWO (2) PAD LOCKS, AND INTERLOCKED WITH COVER IN "CLOSED" POSITION. ENCLOSURES SHALL COMPLY WITH NEMA KS 1; TYPE 1 INDOOR DRY LOCATIONS;
   - B. MOLDED CASE CIRCUIT BREAKER: NEWMAN AB 1, HANDLE LOCKABLE WITH TWO (2) PADLOCKS. FRAME SIZE, TRIP RATING NUMBER OF POLES, AND AUXILIARY DEVICES AS INDICATED. ENCLOSURES SHALL COMPLY WITH NEWMAN AB 1; TYPE 1. INDOOR DRY LOCATIONS. CIRCUIT BREAKERS SHALL HAVE A MINIMUM INTERRUPTING CAPACITY AS FOLLOWS:
   - C. MANUFACTURER: SUBJECT TO COMPLIANCE WITH REQUIREMENTS, PROVIDE PRODUCTS AS SUITABLE FOR INSTALLATION IN EXISTING PANELS. DISCONNECTS SHALL BE BY SQUARE–D.

STANDARD MOUNTING HEIGHTS

(UNLESS OTHERWISE NOTED)

9" below finish ceiling	wall–mounted clocks, program bells, (or as shown on architcctural details)
6" above fire hose cabinet	blue signal light
10'–0"	battery lighting units and remote wall mounted light heads (or 1'–0" below finished ceiling of top of unit).
8'–6"	pendant–hung industrial and strip lighting fixtures.
centered above door or window opening	warning and signal fixtures/signs.
6'–8"–to bottom of device	or 6" below finished ceiling whichever is lower. fire alarm signal devices and illuminated fire signals
6'–6"	top of flush and surface mounted electrical lighting or power panelboards and telephone cabinets.
6'–3"	top of back–mounted wall exit fixtures not mounted above doors).
6'–0"	top of highest electrical safety disconnect switches, magnetic starters, contactors.
3'–8"	wall mounted electrical device, lighting switches, manual motor starters, thermostats, wall–mounted wiremold and g.f.i. receptacles in toilet rooms or for separate sinks (maximum side reach height ADA)
3'–6"	fire alarm pull stations, ADA compliant device rough in
18"	electrical recEQtacles, telEQhone & data outlets
00"	finished floor

MOUNTING HEIGHT NOTES:

1. STANDARD MOUNTING HEIGHTS: (COORDINATE WITH ARCH DRAWINGS) ALL MOUNTING HEIGHTS SHALL BE AS INDICATED BY ARCHITECT. IF NOT INDICATED BY ARCHITECT THEN PROVIDE AS NOTED ABOVE.
2. MOUNTING HEIGHTS TO CENTER OF OUTLETS UNLESS OTHERWISE NOTED. IN MASONRY CONSTRUCTION THE ABOVE MOUNTING HEIGHTS SHALL BE USED FOR REFERENCE TO NEAREST BLOCK OR BRICK COURSING.
3. THE ABOVE MOUNTING HEIGHT SHALL BE ADHERED TO UNLESS SPECIFICALLY NOTED OR DETAILED OTHERWISE ON THE DRAWING OR SPECIFICATIONS.
4. INDICATION (+) NEXT TO A DEVICE INDICATES THAT DEVICE IS MOUNTED ABOVE A COUNTER OR CASEWORK. COORDINATE WITH ARCHITECTURAL DETAIL AND CASEWORK CONTRACTOR.
5. 3'–6" FOR ADA COMPLIANT DEVICES VERIFY EXACT HEIGHT PRIOR TO ROUGH IN.

ELECTRICAL NOTES:

1. WIRING METHODS: WIRING SHALL CONSIST NO ROME (NON–METALLIC) OR CABLES AND WIRES INSTALLED IN RGS CONDUIT, EMT, LIQUIDATING FLEXIBLE METAL CONDUIT AND MC CABLES. RACEWAYS SHALL BE CONCEALED WITHIN FINISHED WALLS AND CEILINGS UNLESS OTHERWISE INDICATED. OTHER THAN ROOF PENETRATIONS REQUIRED TO FEED ROOF MOUNTED EQUIPMENT, RACEWAYS WILL NOT BE ROUTED EXPOSED OUTSIDE OF THE BUILDING.

2. RACEWAYS: RACEWAYS SHALL BE PROVIDED WHERE INDICATED AND REQUIRED AND SHALL BE INSTALLED AS SPECIFIED BELOW, UNLESS OTHERWISE INDICATED. MINIMUM RACEWAY SIZE SHALL BE 3/4 IN. RGS CONDUIT SHALL BE USED FOR ALL OUTDOOR INSTALLATIONS. EMT SHALL BE USED FOR ALL INDOOR INSTALLATIONS. LIQUIDATING FLEXIBLE METAL CONDUIT, 6 FT MAXIMUM LENGTH,SHALL BE USED FOR ALL CONDUIT TERMINATIONS AT EQUIPMENT SUBJECT TO VIBRATION. BUSHINGS, MANUFACTURED FITTINGS, OR BOXES PROVIDING EQUIVALENT MEANS OF PROTECTION SHALL BE INSTALLED ON THE ENDS OF ALL CONDUITS AND SHALL BE OF THE INSULATING TYPE WHERE REQUIRED BY THE N.E.C. ONLY LISTED ADAPTERS SHALL BE USED TO CONNECT EMT TO RGS CONDUIT AND CAST METAL BOXES AND CONDUIT BODIES. PENETRATIONS OF SLABS AND FIRE RATED WALLS SHALL BE FIRESTONE.

   KEEP RACEWAYS AT LEAST 6 IN. AWAY FROM PARALLEL RUNS OF FLUES AND STEAM OR HOT WATER PIPING. INSTALL HORIZONTAL RACEWAY RUNS HIGHER THAN WATER AND STEAM PIPING. RACEWAYS CROSSING STRUCTURAL EXPANSION JOINT SHALL BE PROVIDED WITH SUITABLE EXPANSION FITTINGS OR OTHER SUITABLE MEANS TO COMPENSATE FOR THE BUILDING EXPANSION AND CONTRACTION. RACEWAYS SHALL BE INSTALLED PARALLEL OR PERPENDICULAR TO WALLS, STRUCTURAL MEMBERS AND FEATURES, MECHANICAL DUCT AND PIPING SYSTEMS, OR INTERSECTIONS OF VERTICAL PLANES AND CEILINGS. CHANGES IN DIRECTION OF RUNS SHALL BE ACCOMPLISHED WITH SYMMETRICAL BENDS OR CAST METAL FITTINGS. FILED–MADE ELBOWS AND OFFSETS SHALL BE MADE WITH AN APPROVED HICKEY OR CONDUIT BENDING MACHINE. CRUSHED OR DEFORMED RACEWAY SHALL NOT BE INSTALLED. CARE SHALL BE TAKEN TO PREVENT THE LODGMENT OF DIRT, AND CONSTRUCTION MATERIALS AND DEBRIS IN RACEWAYS DURING THE COURSE OF CONSTRUCTION. CLOGGED RACEWAYS SHALL BE ENTIRELY FREED OF OBSTRUCTIONS OR SHALL BE REPLACED.

   RGS CONDUIT AND EMT SHALL BE SECURELY AND RIGIDLY FASTENED IN PLACE AT INTERVALS OF NOT MORE THAN 10 FT AND WITHIN 3 FT OF FITTINGS AND BOXES WITH APPROVED PIPE STRAPS, WALL BRACKETS, CONDUIT CLAMPS, CONDUIT HANGERS, THREADED C–CLAMPS, OR CEILING TRAPEZE. C–CLAMPS OR BEAM CLAMPS SHALL HAVE STRAP OR ROD TYPE RETAINERS. LOADS AN SUPPORTS SHALL BE COORDINATED WITH SUPPORTING STRUCTURES TO PREVENT DEFORMATION OR DAMAGE TO STRUCTURES, BUT NO LOAD SHALL BE APPLIED TO JOIST BRIDGING. FASTENINGS SHALL BE BY WOOD SCREWS OR SCREW TYPE NAILS TO WOOD; BY TOGGLE BOLTS ON HOLLOW CMU; BY EXPANSION BOLTS ON CONCRETE OR BRICK; AND BY MACHINE SCREWS, WELDED THREADED STUDS, HEAT TREATED OR SPRING STEEL TENSION CLAMPS ON STEEL WORK. NAIL TYPE NYLON ANCHORS OR THREADED STUDS DRIVEN INBY A POWDER CHARGE AND PROVIDED WITH LOCK WASHERS AND NUTS MAY BE USED IN LIEU OF EXPANSION BOLTS OR MACHINE SCREWS. RACEWAYS OR PIPE STRAPS SHALL NOT BE WELDED TO STEEL STRUCTURES. IN PARTITIONS OF LIGHT STEEL CONSTRUCTION, SHEET METAL SCREWS MAY BE USED. CONDUIT SHALL NOT BE SUPPORTED USING WIRE OR NYLON TIES. RACEWAYS SHALL BE INSTALLED AS A COMPLETE SYSTEM AND BE INDEPENDENTLY SUPPORTED FROM THE STRUCTURE. UPPER RACEWAYS SHALL NOT BE THE SUPPORT OF LOWER RACEWAYS. SUPPORTING MAINS WILL NOT BE SHARED BETWEEN ELECTRICAL RACEWAYS AND MECHANICAL DUCTS OR PIPING. MOUNTING HARDWARE SHALL NOT PRESENT SHARP EDGES WHERE PERSONNEL CONTACT IS POSSIBLE. IN MECHANICAL SPACES, "MINERALS" SUPPORTS SHALL NOT BE USED BELOW 10' A.F.F.

3. BOXES: BOXES SHALL BE PROVIDE IN RACEWAY SYSTEMS WHEREVER REQUIRED FOR PULLING OF WIRES, MAKING CONNECTIONS, AND MOUNTING OF DEVICES OR LIGHTING FIXTURES. IN GENERAL, BOXES SHALL BE CONSTRUCTED OF HOT–DIPPED GALVANIZED FINISHED SHEET STEEL. BOXES FOR METALLIC RACEWAYS, 4 IN. BY 4 IN. NOMINAL SIZE AND SMALLER, SHALL BE OF THE CAST METAL HUB TYPE AND CASKETED WHEN LOCATED OUTSIDE OF THE BUILDING. BOXES SHALL BE LISTED AS SUITABLE FOR THE ENVIRONMENTAL CONDITIONS OF THE LOCATION THEY ARE INSTALLED. BOXES FOR MOUNTING OF LIGHTING FIXTURES SHALL BE NOT LESS THAN 4 IN. SQUARE EXCEPT SMALLER BOXES SHALL BE INSTALLED WHERE REQUIRED BY FIXTURE CONFIGURATION. UNLESS OTHERWISE INDICATED, DEVICE BOXES FOR RECEPTACLES SHALL BE MOUNTED WITH THE CENTER OF THE DEVICE BOX APPROXIMATELY 18 IN. AFF. INDICATED, DEVICE BOXES FOR TOGGLE SWITCHES SHALL BE MOUNTED WITH THE CENTER OF THE DEVICE BOX APPROXIMATELY 48 IN. AFF. BOXES AND BOX SUPPORTS SHALL BE FASTENED TO WOOD WITH WOOD SCREWS OR SCREW TYPE NAILS OF EQUAL HOLDING STRENGTH, WITH BOLTS AND METAL EXPANSION SHIELDS ON CONCRETE AND BRICK, WITH TOGGLE BOLTS ON HOLLOW CMU, AND WITH MACHINE SCREWS OR WELDED STUDS ON STEEL WORK. THREADED STUDS DRIVEN IN BY POWDER CHARGE AND PROVIDED WITH LOCKWASHERS AND NUTS, OR NAIL TYPE NYLON ANCHORS MAY BE USED IN LIEU OF EXPANSION SHIELDS OR MACHINE SCREWS. HANGERS SHALL NOT BE FASTENED OR SUPPORTED FROM JOIST BRIDGING.

4. WIRES AND CABLES: EXAMINE RACEWAYS AND BUILDING FINISHES TO RECEIVE WIRES AND CABLES FOR COMPLIANCE WITH INSTALLATION TOLERANCES AND OTHER CONDITIONS. DO NOT PROCEED WITH INSTALLATION UNTIL UNSATISFACTORY CONDITIONS HAVE BEEN CORRECTED. PULL WIRES AND CABLES INTO RACEWAY SIMULTANEOUSLY WHERE MORE THAN ONE IS BEING INSTALLED IN THE SAME RACEWAY. USE PULLING COMPOUND OR LUBRICANT WHERE NECESSARY; COMPOUND USED MUST NOT DETERIORATE CONDUCTORS OR INSULATION. USE PULLING MEANS, INCLUDING FISH TAPE, CABLE, ROPE, AND BASKET–WEAVE WIRE/CABLE GRIPS THAT WILL NOT DAMAGE WIRES/CABLES OR RACEWAY. INSTALL EXPOSED CABLE, PARALLEL AND PERPENDICULAR TO WALLS, STRUCTURAL MEMBERS AND FEATURES, MECHANICAL DUCT AND PIPING SYSTEMS, OR INTERSECTIONS OF VERTICAL PLANES AND CEILING. HORIZONTAL RUNS OF MC CABLE SHALL BE SUPPORTED ON 3 FT MAXIMUM CENTERS. VERTICAL RUNS OF MC CABLES SHALL BE SUPPORTED ON 6 FT MAXIMUM CENTERS EXPOSED PLENUM CABLE SHALL BE SUPPORTED ON 3 FT MAXIMUM CENTERS. THE NUMBER OF SPLICES SHALL BE KEPT TO AN ABSOLUTE MINIMUM. WIRING AT EACH OUTLET SHALL BE INSTALLED WITH AT LEAST 8 IN. SLACK.

5. WIRING DEVICES: INSTALL WIRING DEVICES WHERE INDICATED IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED INSTALLATION, INSTRUCTION APPLICABLE REQUIREMENTS OF THE NEC, AND RECOGNIZED INDUSTRY PRACTICES. INSTALL WIRING DEVICES IN DEVICE BOXES WHICH ARE CLEAN AND FREE FROM DIRT AND CONSTRUCTION MATERIALS AND DEBRIS. INSTALL WIRING DEVICES AFTER WIRING WORK IS COMPLETE. INSTALL DEVICE PLATES AFTER PAINTING WORK IS COMPLETE. TIGHTEN CONNECTORS AND TERMINALS, INCLUDING SCREWS AND BOLTS, IN ACCORDANCE WITH MANUFACTURER;S PUBLISHED TORQUE–TIGHTENING VALUES. PRIOR TO ENERGIZING CIRCUITS, TEST WIRING FOR ELECTRICAL CONTINUITY AND SHORTS. ENSURE PROPER POLARITY OF CONNECTIONS IS MAINTAINED. SUBSEQUENT TO EMERGING, TEST WIRING DEVICES AND DEMONSTRATE COMPLIANCE WITH REQUIREMENTS.

6. GROUNDING: ELECTRICAL SYSTEMS AND EQUIPMENT, METALLIC RACEWAYS AND BOXES, CABLE SHIELDS, METALLIC CABLE SHEATHS AND ARMOR, AND OTHER NON–CURRENT CARRYING METALLIC PARTS OF EQUIPMENT SHALL BE GROUNDED IN CONFORMANCE WITH THE NEC. EQUIPMENT GROUND CONDUCTORS SHALL COMPLY WITH NEC ARTICLE 250 FOR SIZES AND QUANTITIES, EXCEPT WHERE LARGER SIZES AND/OR MORE CONDUCTORS ARE INDICATED. PROVIDE SEPARATE INSULATED GROUND CONDUCTOR IN ALL RACEWAYS, REGARDLESS OF RACEWAY TYPE. SEPARATELY DERIVED SYSTEMS AS DEFINED BY THE NEC SHALL BE GROUNDED IN CONFORMANCE WITH NEC ARTICLE 250 PARA. 26. TERMINATE EQUIPMENT GROUND WIRES FOR FEEDERS AND BRANCH CIRCUITS WITH PRESSURE TYPE GROUND LUGS. WHERE METALLIC CONDUITS TERMINATE AT METALLIC HOUSINGS WITHOUT MECHANICAL AND ELECTRICAL CONNECTION TO HOUSING, TERMINATE EACH CONDUIT WITH A GROUNDING BUSHING. CONNECT GROUNDING BUSHINGS WITH A GROUND WIRE TO THE GROUND BUS IN THE HOUSING. BOND ELECTRICALLY NONCONTIGUOUS CONDUITS AT BOTH ENTRANCES AND EXITS WITH GROUNDING BUSHINGS AND GROUND WIRES. TIGHTEN GROUNDING AND BONDING CONNECTORS AND TERMINALS, INCLUDING SCREWS AND BOLTS, IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED TORQUE–TIGHTENING VALUES.

7. PANELBOARDS: INSTALL PANELBOARDS AND ACCESSORY ITEMS IN ACCORDANCE WITH NEMA PB 1.1, "GENERAL INSTRUCTIONS FOR PROPER INSTALLATION, OPERATION, AND MAINTENANCE OF PANELBOARDS RATED 600 VOLTS OR LESS,"AND MANUFACTURER'S PUBLISHED INSTALLATION INSTRUCTIONS. MOUNT PANELBOARDS PLUMB AND RIGID WITHOUT DISTORTION OF BOX AND WITH THE TOP OF THE TRIM AT 78 IN. AFF UNLESS OTHERWISE INDICATED. PROVIDE NEATLY TYPED AND ACCURATE CIRCUIT DIRECTORIES IN EACH PANELBOARD, REFLECTIVE OF FINAL CIRCUIT CONFIGURATION. PROVIDE FILLER PLATES IN ALL UNUSED SPACES. TRAIN WIRES IN PANELBOARDS GUTTERS NEATLY IN GROUPS, BUNDLE, AND WRAP WITH WIRE TIES. GROUND PANELBOARD IN CONFORMANCE WITH THE NEC. TIGHTEN ELECTRICAL CONNECTORS AND TERMINALS, INCLUDING GROUNDING CONNECTIONS, IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED TORQUE–TIGHTENING VALUES. PERFORM INSULATION RESISTANCE TESTS OF PANELBOARD BUSES, COMPONENTS, AND FEEDER AND BRANCH CIRCUIT WIRING; INSULATION RESISTANCE LESS THAN 100 MEGOHM IS UNACCEPTABLE.

8. DISCONNECTS AND CIRCUIT BREAKERS: PROVIDE DISCONNECTS AND CIRCUIT BREAKERS WHERE INDICATED ON THE ELECTRICAL DRAWINGS AND/OR WHERE REQUIRED BY THE NEC, WHETHER INDICATED ON THE ELECTRICAL DRAWINGS OR NOT. INSTALL DISCONNECTS AND CIRCUIT BREAKERS PLUMB AND LEVEL AND IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED INSTALLATION INSTRUCTIONS. UPON COMPLETION OF INSTALLATION OF DISCONNECTS AND CIRCUIT BREAKERS, ENERGIZE CIRCUITS AND DEMONSTRATE CAPABILITY AND COMPLIANCE WITH REQUIREMENTS. EXCEPT AS OTHERWISE INDICATED, DO NOT DEMONSTRATE DISCONNECTS AND CIRCUIT BREAKERS BY OPERATING THEM UNDER LOAD; HOWEVER, DEMONSTRATE DISCONNECT AND CIRCUIT BREAKERS OPERATION THROUGH SIX OPENING/CLOSING CYCLES WITH CIRCUIT UNLOADED. OPEN DISCONNECT AND CIRCUIT BREAKER ENCLOSURES FOR INSPECTION OF INTERIOR, MECHANICAL AND ELECTRICAL CONNECTIONS, FUSE INSTALLATION IF APPLICABLE, AND FOR VERIFICATION OF TYPE AND RATING OF FUSES INSTALLED IN APPLICABLE. CORRECT DEFICIENCIES THEN RETEST TO DEMONSTRATE COMPLIANCE WITH REQUIREMENTS. REMOVE AND REPLACE DEFECTIVE UNITS WITH NEW UNITS AND RETEST.

9. LIGHTING FIXTURES: INSTALL FIXTURES WHERE, AND AT HEIGHTS, INDICATED IN ACCORDANCE WITH FIXTURE MANUFACTURER'S PUBLISHED INSTALLATION INSTRUCTIONS, REQUIREMENTS OF THE NEC, AND RECOGNIZED INDUSTRY PRACTICES. TIGHTEN ELECTRICAL CONNECTORS AND TERMINALS, INCLUDING GROUND CONNECTIONS, IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED TORQUE–TIGHTENING VALUES. RECESSED OR SEMI RECESSED FIXTURES MAY BE SUPPORTED BY CEILING SUPPORT SYSTEM. INSTALL CEILING SUPPORT SYSTEM RODS OR WIRES AT A MINIMUM OF FOUR (4) RODS OR WIRES PER FIXTURE LOCATED NOT MORE THAN 6 IN. FROM FIXTURE CORNERS. FIXTURES WHICH ARE SMALLER THAN CEILING GRID SHALL BE CENTERED IN ACOUSTICAL CEILING PANEL AND SUPPORTED BY AT LEAST TWO (2) 3/4 IN. METAL CHANNELS SPANNING AND SECURED TO CEILING SYSTEM GRID Tees. FIXTURES WHICH LAY–IN CEILING GRID SYSTEM SHALL BE SECURED IN PLACE BY INSTALLATION OF CLIPS WHICH SECURELY FASTEN FIXTURE TO CEILING GRID Tees. SUPPORT SURFACE MOUNT FIXTURES GREATER THAN 2 FT IN LENGTH AT A POINT IN ADDITION TO THE OUTLET BOX FIXTURE STUD. UPON COMPLETION OF INSTALLATION AND JUST PRIOR TO DEMONSTRATION CLEAN AND RELAMP FIXTURES. LAMP FIXTURES WITH SPECIFIED LAMPS IN ACCORDANCE WITH MANUFACTURER'S PUBLISHED INSTRUCTIONS. UPON COMPLETION OF INSTALLATION, DEMONSTRATE CAPABILITY AND COMPLIANCE WITH REQUIREMENTS. WHERE POSSIBLE, CORRECT MALFUNCTIONING FIXTURES AT THE SITE, THEN RETEST TO DEMONSTRATE COMPLIANCE; OTHERWISE, REMOVE MALFUNCTIONING UNITS AND REPLACE WITH NEW UNITS AND PROCEED WITH RETESTING.

OWNER SIGNATURE

REVISION STATUS

Rev	DATE	DESCRIPTION	DRAWN BY
00	12-10-2023	ISSUED FOR APPROVAL	A.S
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CLIENT/CONSULTANT APPROVAL STATUS

APPROVED	(A)
APPROVED WITH NOTES	(B)
RESUBMIT	(C)
DISAPPROVED	(D)

OWNER:

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DRAWING STATUS:

DISCIPLINE: ELECTRICAL

PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: NOTES

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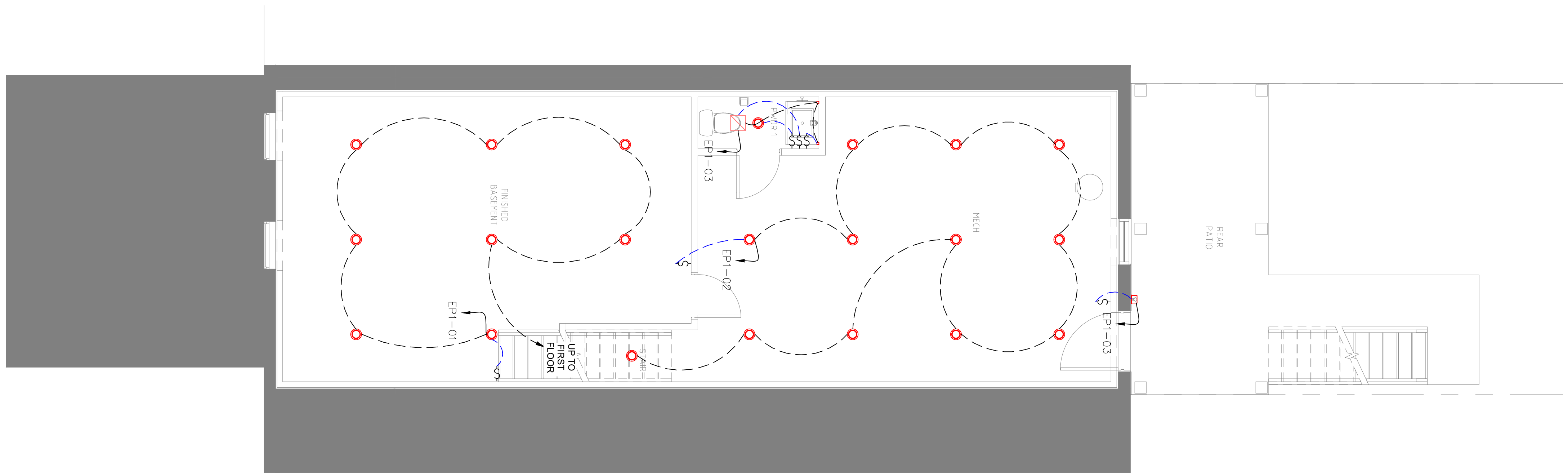
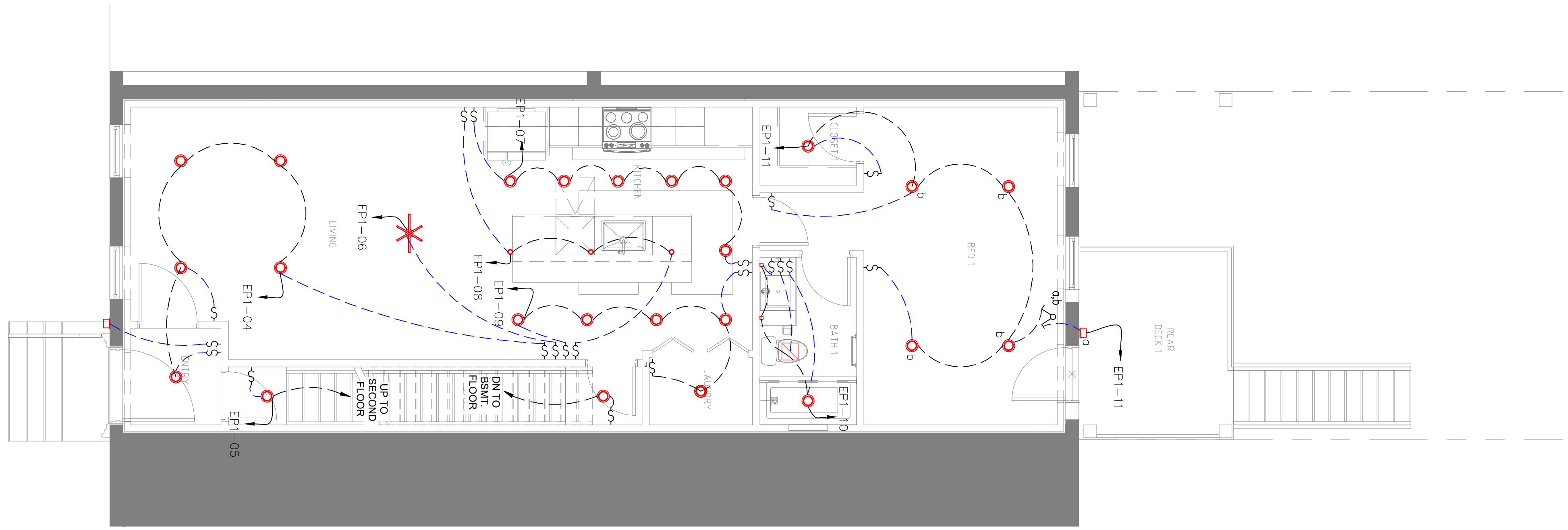
A3



EL 02 **BASEMENT FLOOR LIGHTING PLAN**
SCALE 1/8"=1'-0"



EL 02 **FIRST FLOOR LIGHTING PLAN**
SCALE 1/8"=1'-0"

ELECTRICAL SWITCHING PLAN LEGEND

$	SWITCH
$J	JAMB SWITCH
⌒	SWITCHING CONNECTION

ELECTRICAL SWITCHING GENERAL NOTES

1. IT IS THE CONTRACTOR'S RESPONSIBILITY TO VERIFY EXISTING CONDITIONS AND HOW IT WILL BE AFFECTED BY NEW WORK.
2. ELECTRICAL SWITCHING LOCATIONS ARE SHOWN FOR REFERNCE ONLY.
3. FINAL ELECTRICAL DEVICE AND EQUIPMENT LOCATIONS TO COMPLY WITH CODE REQUIREMENTS.
4. COORDINATE EXISTING AND NEW APPLIANCE AND EQUIPMENT WITH FINAL ELECTRICAL DEVICE LOCATIONS.
5. REFER TO REFLECTED CEILING PLANS FOR LIGHT FIXTURE TYPES.

OWNER SIGNATURE

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PROJECT TITLE: 23-5550F_NCARROLLTON636

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JOB NO 632-021	SHT NO EL-02	SCALE: 1/8"=1'-0" @A3

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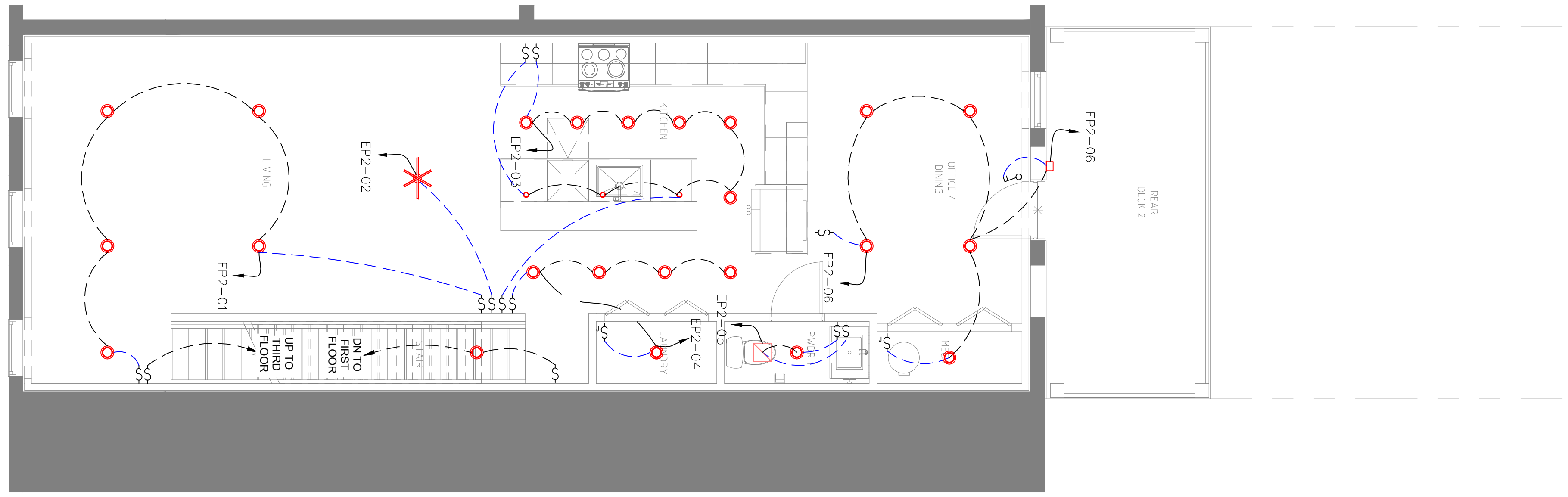
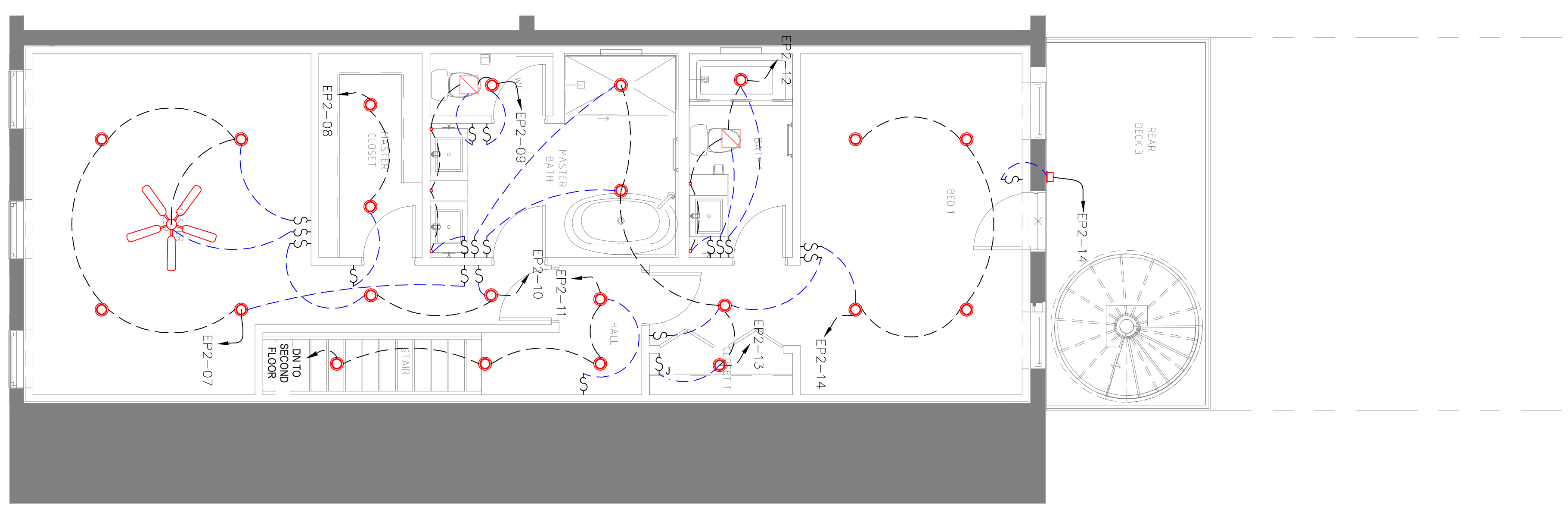


EL 03 SECOND FLOOR LIGHTING PLAN
SCALE 1/8"=1'-0"



EL 03 THIRD FLOOR LIGHTING PLAN
SCALE 1/8"=1'-0"

ELECTRICAL SWITCHING PLAN LEGEND

$	SWITCH
$J	JAMB SWITCH
⌒	SWITCHING CONNECTION

ELECTRICAL SWITCHING GENERAL NOTES

1. IT IS THE CONTRACTOR'S RESPONSIBILITY TO VERIFY EXISTING CONDITIONS AND HOW IT WILL BE AFFECTED BY NEW WORK.
2. ELECTRICAL SWITCHING LOCATIONS ARE SHOWN FOR REFERNCE ONLY.
3. FINAL ELECTRICAL DEVICE AND EQUIPMENT LOCATIONS TO COMPLY WITH CODE REQUIREMENTS.
4. COORDINATE EXISTING AND NEW APPLIANCE AND EQUIPMENT WITH FINAL ELECTRICAL DEVICE LOCATIONS.
5. REFER TO REFLECTED CEILING PLANS FOR LIGHT FIXTURE TYPES.

OWNER SIGNATURE

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DISAPPROVED	(D)

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DRAWING STATUS:

DISCIPLINE:
ELECTRICAL

PROJECT TITLE:
23-5550F_NCARROLLTON636

DRAWING TITLE:
SECOND & THIRD FLOOR LIGHTING PLANS

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JOB NO 632-021	SHT NO EL-03	SCALE: 1/8"=1'-0" @A3

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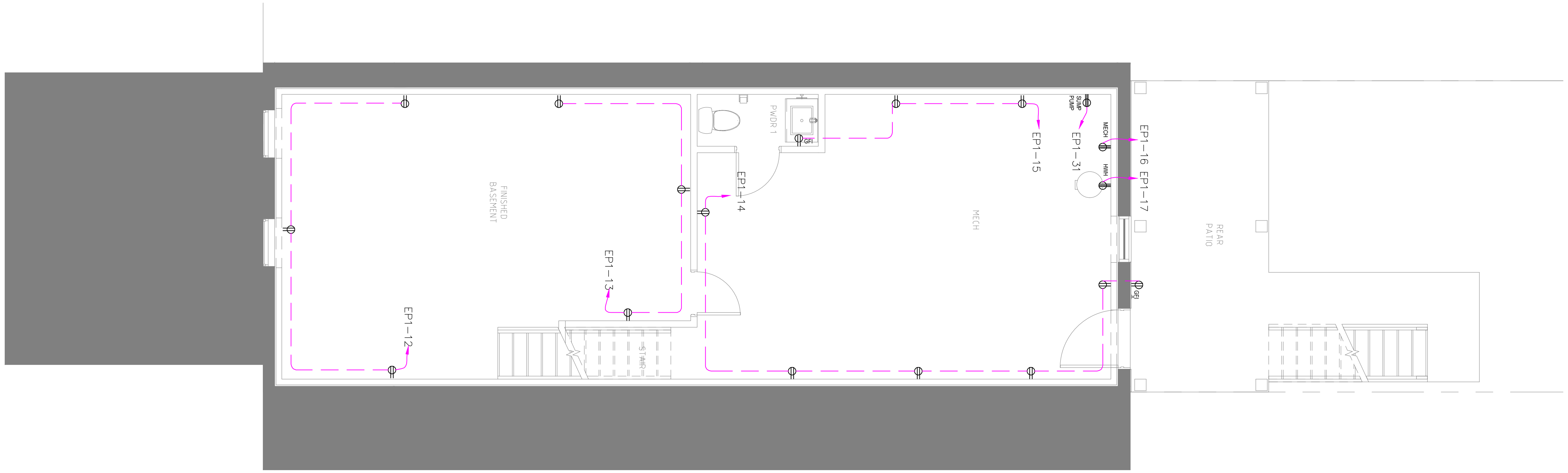
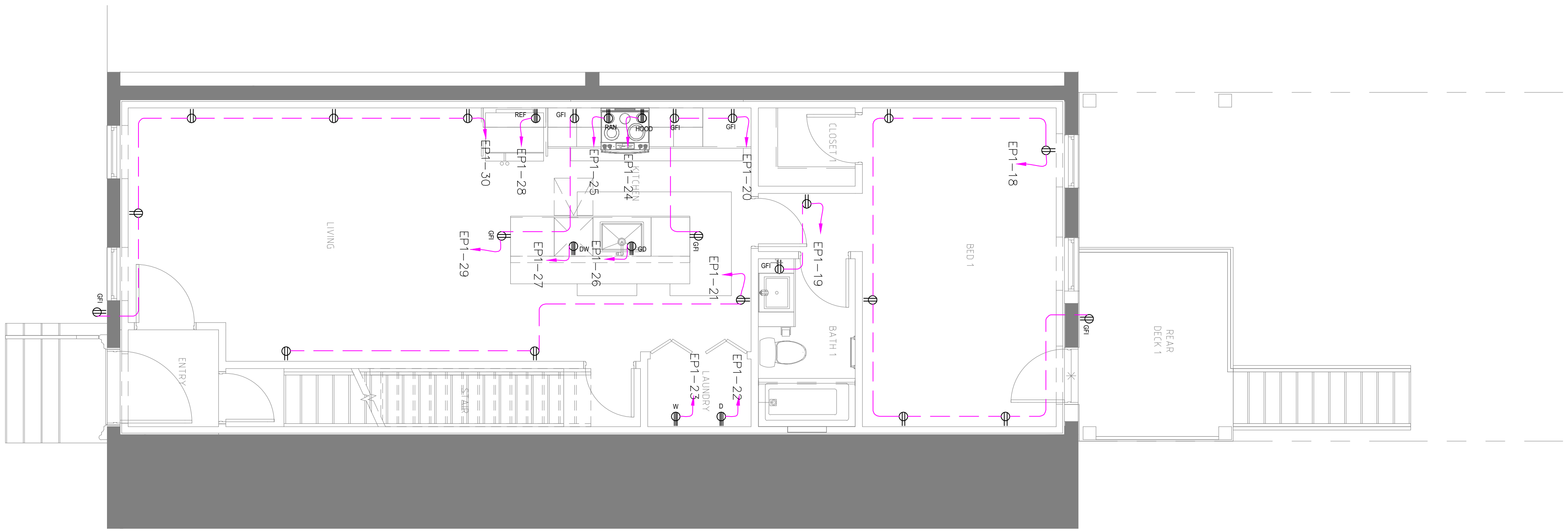


EL 04 **BASEMENT FLOOR POWER PLAN**
SCALE 1/8"=1'-0"



EL 04 **FIRST FLOOR POWER PLAN**
SCALE 1/8"=1'-0"

ELECTRICAL POWER PLAN LEGEND

(symbol)	DUPLEX OUTLET
GFI	GFI OUTLET
(symbol)	APPLIANCE OUTLET
P#	ELECTRICAL POWER KEY NOTE
(shaded)	SHADED AREA NOT IN CONTRACT (NIC)

ELECTRICAL POWER GENERAL NOTES

1. IT IS THE CONTRACTOR'S RESPONSIBILITY TO VERIFY EXISTING CONDITIONS AND HOW IT WILL BE AFFECTED BY NEW WORK.
2. ELECTRICAL POWER LOCATIONS ARE SHOWN FOR REFERNCE ONLY.
3. FINAL ELECTRICAL DEVICE AND EQUIPMENT LOCATIONS TO COMPLY WITH CODE REQUIREMENTS.
4. COORDINATE EXISTING AND NEW APPLIANCE AND EQUIPMENT WITH FINAL ELECTRICAL DEVICE LOCATIONS.

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DISCIPLINE: ELECTRICAL

PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: BASEMENT & FIRST FLOOR POWER PLANS

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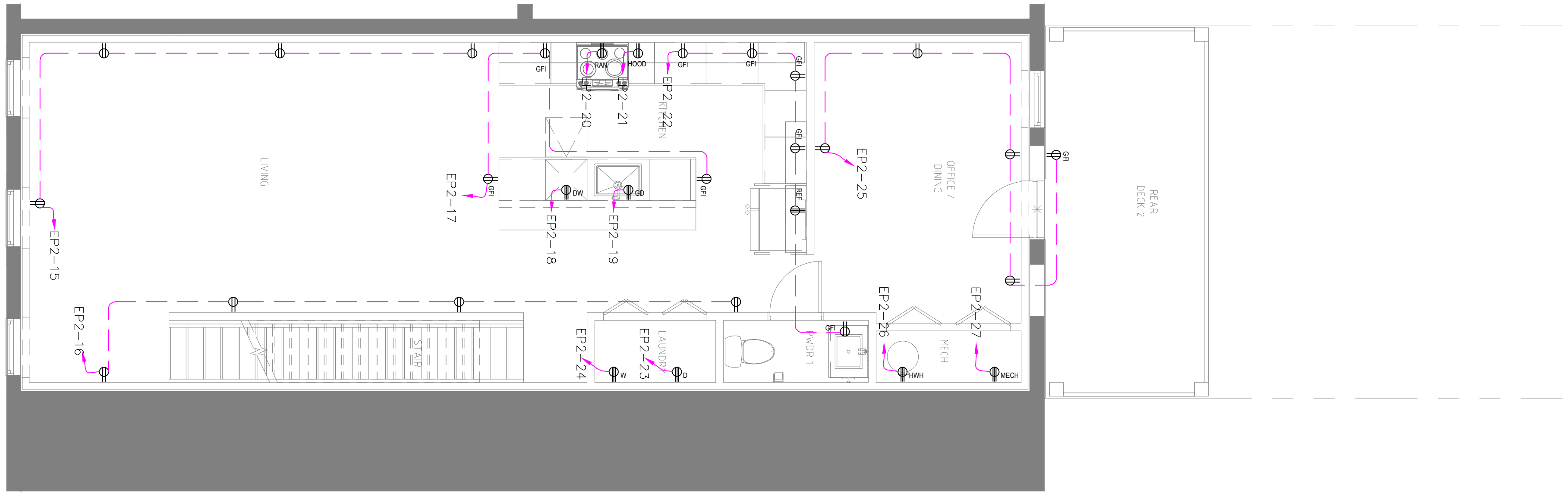
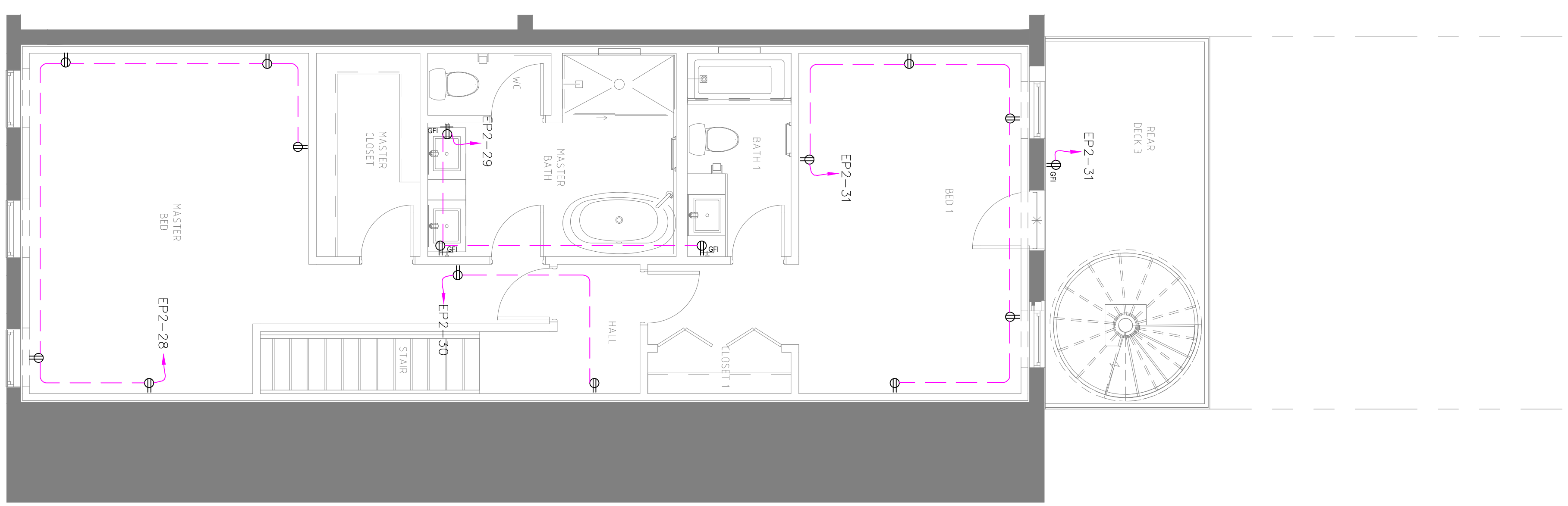
JOB NO	SHT NO	SCALE:
632-021	EL-04	1/8"=1'-0" @A3



EL 05 SECOND FLOOR POWER PLAN
SCALE 1/8"=1'-0"



EL 05 THIRD FLOOR POWER PLAN
SCALE 1/8"=1'-0"

ELECTRICAL POWER PLAN LEGEND

	DUPLEX OUTLET
GFI	GFI OUTLET
	APPLIANCE OUTLET
P#	ELECTRICAL POWER KEY NOTE
	SHADED AREA NOT IN CONTRACT (NIC)

ELECTRICAL POWER GENERAL NOTES

1. IT IS THE CONTRACTOR'S RESPONSIBILITY TO VERIFY EXISTING CONDITIONS AND HOW IT WILL BE AFFECTED BY NEW WORK.
2. ELECTRICAL POWER LOCATIONS ARE SHOWN FOR REFERNCE ONLY.
3. FINAL ELECTRICAL DEVICE AND EQUIPMENT LOCATIONS TO COMPLY WITH CODE REQUIREMENTS.
4. COORDINATE EXISTING AND NEW APPLIANCE AND EQUIPMENT WITH FINAL ELECTRICAL DEVICE LOCATIONS.

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DRAWING STATUS:

DISCIPLINE: ELECTRICAL

PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: SECOND & THIRD FLOOR POWER PLANS

DRAWN BY:	CHECKED BY:	DATE:
A.S	N.B	12 OCT. 2023
JOB NO	SHT NO	SCALE:
632-021	EL-05	1/8"=1'-0" @A3

A3

PANELBOARD SCHEDULE

PANEL TAG: EP-01 | **MAINS:** 200AT/250AF,110/120V,1P,60HZ | NEUTRAL: YES
LOCATION: FLOOR PLAN | **MOUNTING:** SURFACE MOUNTED | GROUND: YES
ENCLOSURE: IP42 | **VOLTAGE:** 110/120V,1P,4W
FEEDER ENTRANCE: TOP

LOAD SERVED	WIRE & CONDUIT SIZE (INCH)	LOAD (VA) ØA	ØB	ØC	CB	POLE	CKT NO.	R	Y	B	CKT NO.	POLE	CB	LOAD (VA) ØA	ØB	ØC	WIRE & CONDUIT SIZE (INCH)	LOAD SERVED
LIGHTING	2#12+1#12G IN 3/4"C	120.00			20	1P	1				2	1P	20	180.00			2#12+1#12G IN 3/4"C	LIGHTING
LIGHTING	2#12+1#12G IN 3/4"C		1545.00		20	1P	3				4	1P	20		150.00		2#12+1#12G IN 3/4"C	LIGHTING
LIGHTING	2#12+1#12G IN 3/4"C			30.00	20	1P	5				6	1P	20			75.00	2#12+1#12G IN 3/4"C	LIGHTING
LIGHTING	2#12+1#12G IN 3/4"C	90.00			20	1P	7				8	1P	20	45.00			2#12+1#12G IN 3/4"C	LIGHTING
LIGHTING	2#12+1#12G IN 3/4"C		75.00		20	1P	9				10	1P	20		1545.00		2#12+1#12G IN 3/4"C	LIGHTING
LIGHTING	2#12+1#12G IN 3/4"C			90.00	20	1P	11				12	1P	20			600.00	2#12+1#12G IN 3/4"C	RECEPTACLE
RECEPTACLE	2#12+1#12G IN 3/4"C	600.00			20	1P	13				14	1P	20	1200.00			2#12+1#12G IN 3/4"C	RECEPTACLE
RECEPTACLE	2#12+1#12G IN 3/4"C		600.00		20	1P	15				16	1P	60		3000.00		2#8+1#10G IN 3/4"C	HVAC
H/W TANK (CONDUCTOR SIZE 10/2)	2#8+1#10G IN 3/4"C			2160.00	50	2P	17				18	1P	20			1200.00	2#12+1#12G IN 3/4"C	RECEPTACLE
RECEPTACLE	2#12+1#12G IN 3/4"C	400.00			20	1P	19				20	1P	20	600.00			2#12+1#12G IN 3/4"C	RECEPTACLE
RECEPTACLE	2#12+1#12G IN 3/4"C		600.00		20	1P	21				22	2P	30		1500.00		2#10+1#12G IN 3/4"C	WASHER
DRYER	2#10+1#12G IN 3/4"C			1000.00	30	2P	23				24	2P	50			1500.00	2#10+1#12G IN 3/4"C	HOOD
RANGE	2#10+1#12G IN 3/4"C	400.00			50	2P	25				26	1P	20	1000.00			2#12+1#12G IN 3/4"C	GARBAGE (AF/GF)
DISHWASHER (AF/GF)	2#10+1#12G IN 3/4"C		1500.00		15	1P	27				28	1P	20		1500.00		2#12+1#12G IN 3/4"C	RECEPTACLE
RECEPTACLE	2#12+1#12G IN 3/4"C			400.00	20	1P	29				30	1P	20			1000.00	2#12+1#12G IN 3/4"C	RECEPTACLE
SUMP PUMP	2#12+1#12G IN 3/4"C	240.00			15	2P	31				32	1P	20					SPARE
SPARE					20	1P	33				34	1P	20					SPARE
		1850.00	4320.00	3680.00										3025.00	7695.00	4375.00		

1

ØA = 4875.00 | TOTAL CONNECTED LOAD = 24945.00 | I (AMP) = 59.22
ØB = 12015.00 | DEMAND FACTOR = 1
ØC = 8055.00 | TOTAL CONNECTED LOAD +25% SPARE= 31181.25

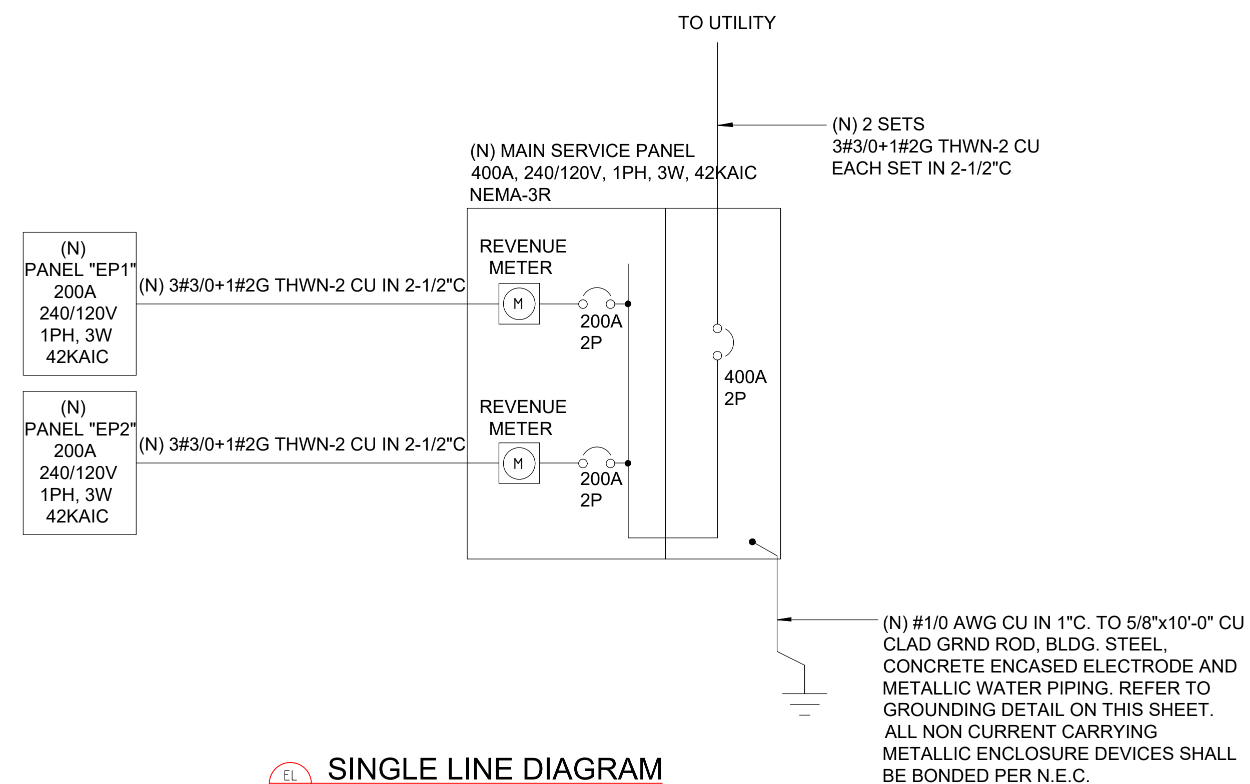
PANELBOARD SCHEDULE

PANEL TAG: EP-02 | **MAINS:** 200AT/250AF,110/120V,1P,60HZ | NEUTRAL: YES
LOCATION: FLOOR PLAN | **MOUNTING:** SURFACE MOUNTED | GROUND: YES
ENCLOSURE: IP42 | **VOLTAGE:** 110/120V,1P,4W
FEEDER ENTRANCE: TOP

LOAD SERVED	WIRE & CONDUIT SIZE (INCH)	LOAD (VA) ØA	ØB	ØC	CB	POLE	CKT NO.	R	Y	B	CKT NO.	POLE	CB	LOAD (VA) ØA	ØB	ØC	WIRE & CONDUIT SIZE (INCH)	LOAD SERVED
LIGHTING	2#12+1#12G IN 3/4"C	75.00			20	1P	1				2	1P	20	75.00			2#12+1#12G IN 3/4"C	LIGHTING
LIGHTING	2#12+1#12G IN 3/4"C		135.00		20	1P	3				4	1P	20		75.00		2#12+1#12G IN 3/4"C	LIGHTING
LIGHTING	2#12+1#12G IN 3/4"C			1515.00	20	1P	5				6	1P	20			90.00	2#12+1#12G IN 3/4"C	LIGHTING
LIGHTING	2#12+1#12G IN 3/4"C	135.00			20	1P	7				8	1P	20	60.00			2#12+1#12G IN 3/4"C	LIGHTING
LIGHTING	2#12+1#12G IN 3/4"C		1560.00		20	1P	9				10	1P	20		60.00		2#12+1#12G IN 3/4"C	LIGHTING
LIGHTING	2#12+1#12G IN 3/4"C			60.00	20	1P	11				12	1P	20			1545.00	2#12+1#12G IN 3/4"C	LIGHTING
LIGHTING	2#12+1#12G IN 3/4"C	60.00			20	1P	13				14	1P	20	90.00			2#12+1#12G IN 3/4"C	LIGHTING
RECEPTACLE	2#12+1#12G IN 3/4"C		800.00		20	1P	15				16	1P	20		800.00		2#12+1#12G IN 3/4"C	RECEPTACLE
RECEPTACLE	2#12+1#12G IN 3/4"C			600.00	20	1P	17				18	1P	15			1500.00	2#10+1#12G IN 3/4"C	DISHWASHER (AF/GF)
GARBAGE (AF/GF)	2#12+1#12G IN 3/4"C	1500.00			20	1P	19				20	2P	50	400.00			2#10+1#12G IN 3/4"C	RAN
HOOD	2#10+1#12G IN 3/4"C		1500.00		50	2P	21				22	1P	20		1200.00		2#12+1#12G IN 3/4"C	RECEPTACLE
WASHER	2#10+1#12G IN 3/4"C			1500.00	30	2P	23				24	2P	30			1000.00	2#10+1#12G IN 3/4"C	DRYER
RECEPTACLE	2#12+1#12G IN 3/4"C	1000.00			20	1P	25				26	2P	50	4500.00			2#8+1#10G IN 3/4"C	H/W TANK (CONDUCTOR SIZE 10/2)
HVAC	2#8+1#10G IN 3/4"C		3000.00		60	1P	27				28	1P	20		1000.00		2#12+1#12G IN 3/4"C	RECEPTACLE
RECEPTACLE	2#12+1#12G IN 3/4"C			600.00	20	1P	29				30	1P	20			400.00	2#12+1#12G IN 3/4"C	RECEPTACLE
RECEPTACLE	2#12+1#12G IN 3/4"C	1200.00			20	2P	31				32	1P	20					SPARE
SPARE					20	1P	33				34	1P	20					SPARE
SPARE					20	1P	35											
		3970.00	6995.00	4275.00										5125.00	3135.00	4535.00		

1

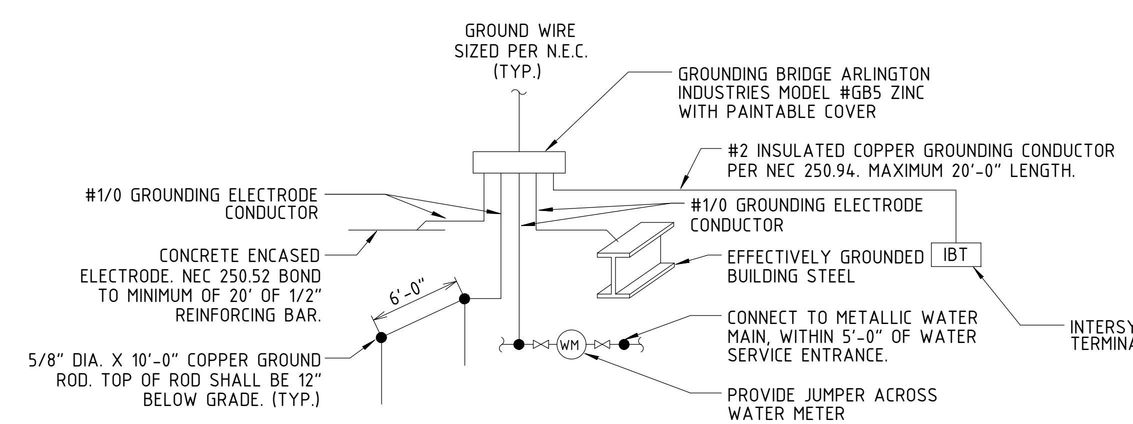
ØA = 9095.00 | TOTAL CONNECTED LOAD = 28035.00 | I (AMP) = 66.56
ØB = 10130.00 | DEMAND FACTOR = 1
ØC = 8810.00 | TOTAL CONNECTED LOAD +25% SPARE= 35043.75



EL 06 SINGLE LINE DIAGRAM
SCALE NTS

SINGLE LINE DIAGRAM GENERAL NOTES:

A. ALL ELECTRICAL SERVICE ENTRANCE EQUIPMENT SHALL BE NEMA-3R RATED IF INSTALLED OUTDOORS.
B. ALL ELECTRICAL EQUIPMENT AND WIRE RATED @ 75ºC CONTINUOUS DUTY.
C. PRIOR TO CONSTRUCTION, THE ELECTRICAL CONTRACTOR SHALL:
   a. VERIFY ALL EXISTING CONDITIONS IN FIELD PRIOR TO CONSTRUCTION.
   b. COORDINATE THE ELECTRICAL SERVICE WITH UTILITY COMPANY REPRESENTATIVE.
   c. NOTIFY THE ENGINEER OF ANY CHANGES REQUIRED TO COMPLETE NEW CONSTRUCTION.
D. ANY NEW ELECTRICAL SERVICE EQUIPMENT SHALL MATCH EXISTING A.I.C. RATING.
E. OVERCURRENT DEVICES OF ENTIRE DISTRIBUTION SYSTEM SHALL MEET STATED FAULT CURRENT VALUES WITH FULLY RATED EQUIPMENT. REFER TO PANEL SCHEDULES FOR ADDITIONAL REQUIREMENTS.
F. WHERE A DISCREPANCY EXISTS BETWEEN EQUIPMENT ON THE SINGLE LINE DIAGRAM AND THE DETAILED SCHEDULES, THE ITEM OR ARRANGEMENT WITH BETTER QUALITY, GREATER QUANTITY, OR HIGHER COST SHALL BE USED.
G. ALL DISCREPANCIES SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER.
H. REFER TO SINGLE CONDUCTOR FEEDER AND BRANCH CIRCUIT SCHEDULE ON SHEET THIS SHEET FOR WIRE SIZES WHERE NOT INDICATED. ELECTRICAL CONTRACTOR SHALL SIZE ALL WIRES THAT ARE NOT INDICATED PER C.E.C.
I. ELECTRICAL CONTRACTOR SHALL CONFIRM EXACT FAULT CONTRIBUTION WITH UTILITY COMPANY AND ALL EQUIPMENT RATING SHALL EXCEED FAULT CURRENT PROVIDED BY UTILITY.



EL 06 GROUNDING DETAIL
SCALE NTS

OWNER SIGNATURE

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00	12-10-2023	ISSUED FOR APPROVAL	A.S
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CLIENT/CONSULTANT APPROVAL STATUS	
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RESUBMIT	(C)
DISAPPROVED	(D)

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DRAWING STATUS:

DISCIPLINE: ELECTRICAL

PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: PANEL SCHEDULE & SINGLE LINE DIAGRAM

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A3

GENERAL PLUMBING NOTES:

1. LOCATION AND SIZES OF ALL EXISTING SERVICES ARE APPROXIMATE. VERIFY ALL UTILITIES PRIOR TO COMMENCING WORK.

2. FOR EXACT LOCATION OF PLUMBING FIXTURES, DRAINS AND EQUIPMENT SEE ARCHITECTURAL DRAWINGS.

3. ALL DOMESTIC WATER AND VENT PIPING SHOWN TO BE RUN ABOVE CEILING UNLESS OTHERWISE NOTED.

4. ALL SANITARY PIPING SHOWN TO BE RUN BELOW FLOOR UNLESS OTHERWISE NOTED.

5. ENTIRE INSTALLATION INCLUDING MATERIALS, EQUIPMENT AND WORKMANSHIP SHALL CONFIRM WITH ALL APPLICABLE LAWS, CODES AND REGULATIONS OF MUNICIPAL, STATE AND FEDERAL AUTHORITIES, ALSO THE PLUMBING CODE 2004, INTERNATIONAL BUILDING CODE 2009, AND OTHER REGULATORY BODIES HAVING JURISDICTION OVER THE CLASS OF WORK.

6. THE CONTRACTOR SHALL MAKE TESTS FOR ACCEPTANCE AND APPROVAL AS REQUIRED BY THE CODE. REQUIRED TESTS SHALL BE PERFORMED IN THE PRESENCE OF OWNER'S REPRESENTATIVE UNLESS WAIVED IN WRITING.

7. THE CONTRACTOR SHALL FIELD COORDINATE THE WORK WITH THAT OF ALL OTHER TRADES.

8. ALL DOMESTIC WATER PIPING SHALL BE INSULATED AS REQUIRED.

9. DOMESTIC WATER PIPING MATERIAL: ASTM B88 TYPE L OR M COPPER TUBE WITH WROUGHT OR CAST COPPER FITTINGS AND LEAD-FREE SOLDERED JOINTS. INSTALL DOMESTIC WATER PIPING LEVEL WITHOUT PITCH AND PLUMB.

DRAIN, WASTE, STORM, AND VENT PIPING MATERIAL: FOR BELOW GROUND SOIL, WASTE AND BENT PIPING SHOULD BE SERVICE WEIGHT HUB AND SPIGOT CAST IRON CONFORMING TO ASTM A74, WITH LEAD AND OAKUM JOINTS AS ALLOWED BY PLUMBING CODE;
FOR ABOVE GROUND SOIL, WASTE AND VENT PIPING SHOULD BE HUBLESS CAST-IRON.

10. ANY ITEM OF MATERIALS NOT INDICATED ON THE DRAWINGS OR NOT SPECIFIED AND WHICH ARE REQUIRED FOR THE COMPLETE INSTALLATION OF ANY PART OF THE WORK DESCRIBED, SHALL BE FURNISHED AS CODE REQUIRED.

11. INSTALL CLEANOUT AT OR NEAR THE BOTTOM OF SANITARY/WASTE STACK.

12. PROVIDE PROPER ACCESS PANELS AS SPECIFIED IN WALLS AND CEILINGS FOR ANY EQUIPMENT, DEVICES, VALVES, SHOCK ABSORBERS, TRAP PRIMERS, CLEANOUTS ETC. REQUIRING ACCESS.

		PLUMBING FIXTURE AND FAUCETS REFERENCED STANDARDS
MARK	DESCRIPTION	STANDARD
WC-1	WATER CLOSET	WATER CLOSETS SHALL CONFIRM TO THE WATER CONSUMPTION REQUIREMENTS OF SECTION 604.4 AND SHALL CONFIRM TO ANSI Z124.4. ASME A112.19.2M, CSA B45.1, CSA B45.4 OR CSA B45.5. WATER CLOSETS SHALL CONFIRM TO THE HYDRAULIC PERFORMANCE REQUIREMENTS OF ASME A112.19.19.9M, CSA B45.1, CSA B45.4 OR CSA B45.5. ELECTRO HYDRAULIC WATER CLOSETS SHALL COMPLY WITH ASME A112.19.13.
L-1	LAVATORY	LAVATORIES SHALL CONFIRM TO ANSI Z124.3. ASME A112.19.1M, ASME A112.19.2M, ASME A112.19.3M, ASME A112.19.4M, ASME A112.19.9M, CSA B45.2, CSA B45.3, CSA B45.4, ROUP WASH UP EQUIPMENT SHALL CONFIRM TO THE REQUIREMENTS OF SECTION 402. EVERY 20 INCHES (608MM) OF RIM SPACE SHALL BE CONSIDERED AS ONE LAVATORY.
KS-1	KITCHEN SINK	SINKS SHALL CONFIRM TO ANSI Z124.6, ASME A112.19.1M, ASME A112.19.2M, ASME A112.19.3M, ASME A112.19.4M, ASME A112.19.9M, CSA B45.1, CSA B45.2, CSA B45.3, OR CSA B45.4,
	FAUCETS AND FIXTURE FITTINGS	FAUCETS AND FIXTURE FITTINGS SHALL CONFORM TO ASME A112.18.1 OR CSA B125. FAUCETS AND FIXTURE FITTINGS THAT SUPPLY DRINKING WATER FOR HUMAN INGESTION SHALL CONFORM TO THE REQUIREMENTS OF NSF 61, SECTION 9. FLEXIBLE WATER CONNECTORS EXPOSED TO CONTINUOUS PRESSURE SHALL CONFORM TO THE REQUIREMENTS OF SECTION 605.6 HAND HELD SHOWERS SHALL CONFORM TO ASSE 1014 OR CSA B125. SHOWER AND TUB SHOWER COMBINATION VALVES SHALL BE BALANCED PRESSURE, THERMOSTATIC OR COMBINATION BALANCED PRESSURE/THERMOSTATIC VALVES THAT CONFORM TO THE REQUIREMENTS OF ASSE 1016 OR CSA B125, MULTIPLE (GANG) SHOWERS SUPPLIED WITH A SINGLE TEMPERED WATER SUPPLY PIPE SHALL HAVE THE WATER SUPPLY FOR SUCH SHOWERS CONTROLLED BY A MASTER THERMOSTATIC MIXING VALVE COMPLYING WITH ASSE 1017. SHOWER AND TUB SHOWER COMBINATION VALVES AND MASTER THERMOSTATIC MIXING VALVES REQUIRED BY THIS SECTION SHALL BE EQUIPPED WITH A MEANS TO LIMIT THE MAXIMUM SETTING OF THE VALVE TO 120°F (49°C), WHICH SHALL BE FIELD ADJUSTED IN ACCORDANCE WITH THE MANUFACTURER'S INSTRUCTIONS. FAUCETS AND FIXTURE FITTINGS WITH HOSE CONNECTED OUTLETS SHALL CONFORM TO ASME A112.18.3M. TEMPERATURE ACTUATED, FLOW REDUCTION DEVICES. WHERE INSTALLED FOR INDIVIDUAL FIXTURE FITTINGS, SHALL CONFORM TO ASSE 1062. SUCH VALVES SHALL NOT BE USED ALONE AS A SUBSTITUTE FOR THE BALANCED PRESSURE. THERMOSTATIC OR COMBINATION SHOWER VALVES REQUIRED IN SECTION 424.3

PLUMBING NOTES:

1. ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE NOTED.
2. ALL PLUMBING INSTALLATIONS SHALL CONFIRM WITH THE REQUIREMENTS OF MANUFACTURER RECOMMENDATION AND OR UNIFORM PLUMBING CODE (UPC).
3. ALL PLUMBING MATERIALS SHALL BE AS SPECIFIED IN THE CONTRACT SPECS AND BOQ.
4. ALL PIPE SIZES SHOWN ARE NOMINAL DIAMETER IN INCHES. UNLESS OTHERWISE MENTIONED.
5. INSTALLER SHOULD NOT SCALE THE DRAWING.
6. ALL PIPE PENETRATIONS IN WALL & SLABS SHALL BE THROUGH PIPE SLEEVES.
7. IF ANY ERROR OR OMISSION IS FOUND IN DRAWING THIS OFFICE MAY BE CONSULTED FOR RECTIFICATION.

TABLE 604.4
MAXIMUM FLOW RATES AND CONSUMPTION FOR PLUMBING FIXTURES AND FIXTURE FITTINGS

PLUMBING FIXTURE R FIXTURE FITTING	MAXIMUM FLOW RATE OR QUANTITY b
LAVATORY, PRIVATE	1.5 gpm at 60 psi
SINK FAUCET	1.5 gpm at 60 psi
WATER CLOSET	1.28 GALLONS PER FLUSHING CYCLE

FOR SI: 1 GALLON = 3.785 L, 1 GALLON PER MINUTE = 3.785 L/M, 1 POUND PER SQUARE INCH = 6.895 KPA.

A. A HAND HELD SHOWER SPRAY IS A SHOWER HEAD.
B. CONSUMPTION TOLERANCES SHALL BE DETERMINED FROM REFERENCED STANDARDS.
C. WATER FACTOR IN GALLONS PER CYCLE CUBIC FOOT

ALL PLUMBING FIXTURES SHALL CONTRIBUTE TO A LEEDS SOLUTION AND SHALL MEET EPA CRITERIA AND BEEN CONSIDERED AS WATER EFFICIENT PRODUCTS AS STIPULATED BY THE LEEDS TA 2009 REFERENCE GUIDE (20% REDUCTION REQUIRED).

2010 F.B.C PLUMBING AS PER SECTION 604.9
A WATER HAMMER ARRESTOR SHALL BE INSTALLED WHERE QUICK CLOSING VALVES ARE UTILIZED. WATER HAMMER ARRESTORS SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S SPECIFICATIONS. WATER HAMMER ARRESTORS SHALL CONFORM TO ASSE 1010.

OWNER SIGNATURE

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CLIENT/CONSULTANT APPROVAL STATUS	
APPROVED	(A)
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RESUBMIT	(C)
DISAPPROVED	(D)

OWNER:

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DRAWING STATUS:

DISCIPLINE: PLUMBING

PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: NOTES

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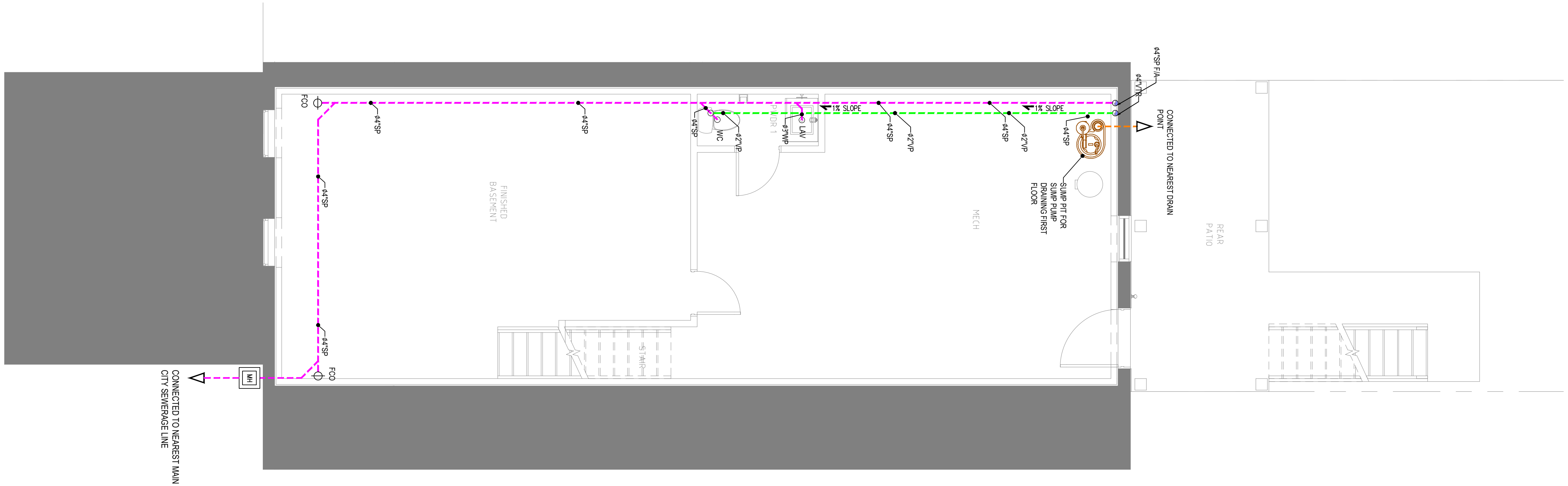
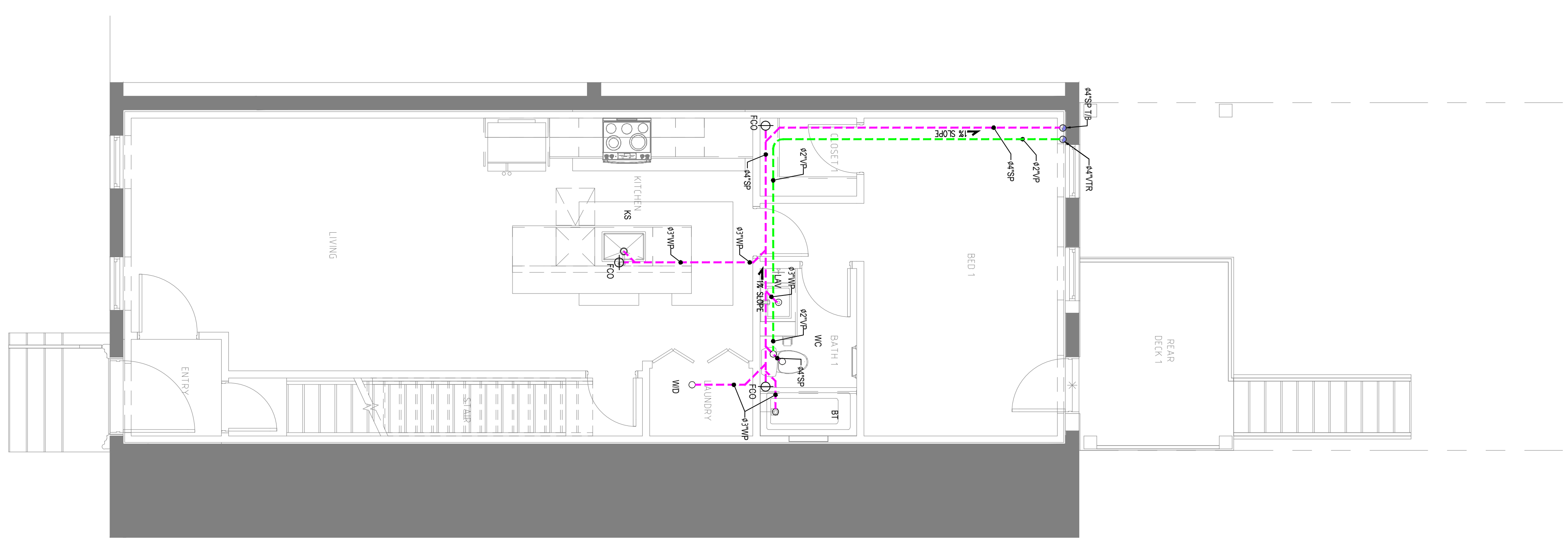


BASEMENT FLOOR DRAINAGE PLAN

PL 02 SCALE 1/8"=1'-0"



FIRST FLOOR DRAINAGE PLAN

PL 02 SCALE 1/8"=1'-0"

PLUMBING LEGEND:

SYMBOL	DESCRIPTION
	HWS - HOT WATER SUPPLY
	CWS - COLD WATER SUPPLY
	VP-VENT PIPE
	SP-SOIL PIPE
	WP-WASTE PIPE
	WP-WASTE PIPE FOR SUMP PUMP
	SHUT OFF VALVE
	GATE VALVE
WH	WATER HEATER
GT	GREASE TRAP
WC	WATER CLOSET
KS	KITCHEN SINK
AS	ABLUTION SPRAY
FCO	FLOOR CLEANOUT
FD	FLOOR DRAIN
LAV	LAVATORY

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RESUBMIT	(C)
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DRAWING STATUS:

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PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: BASEMENT & FIRST FLOOR DRAINAGE PLANS

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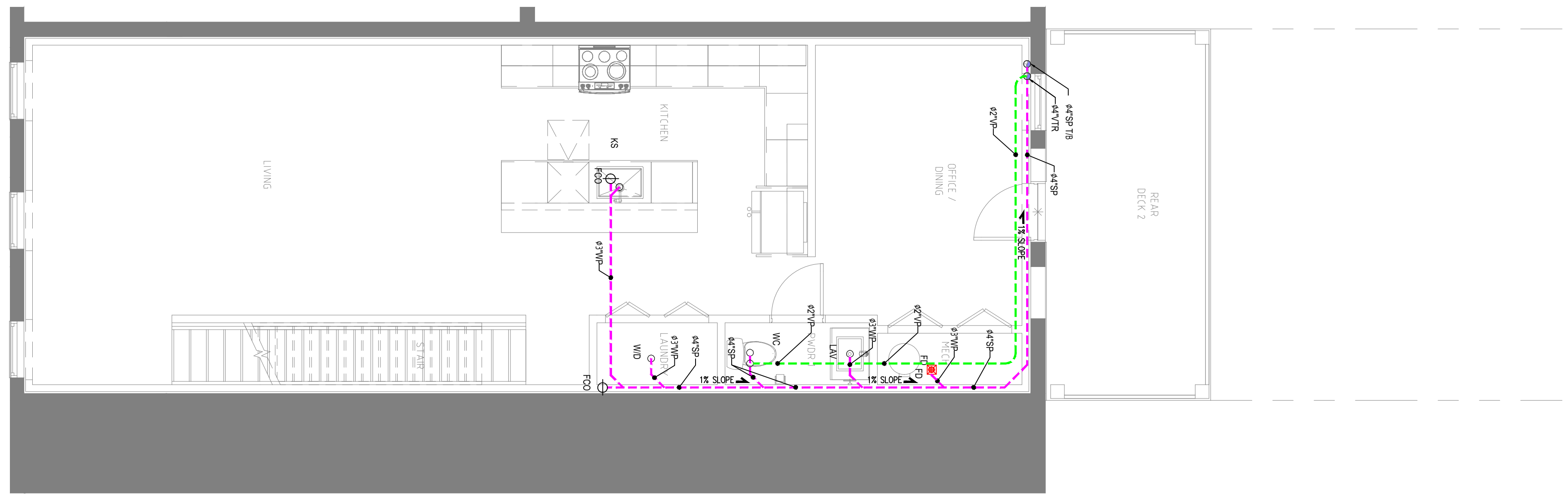
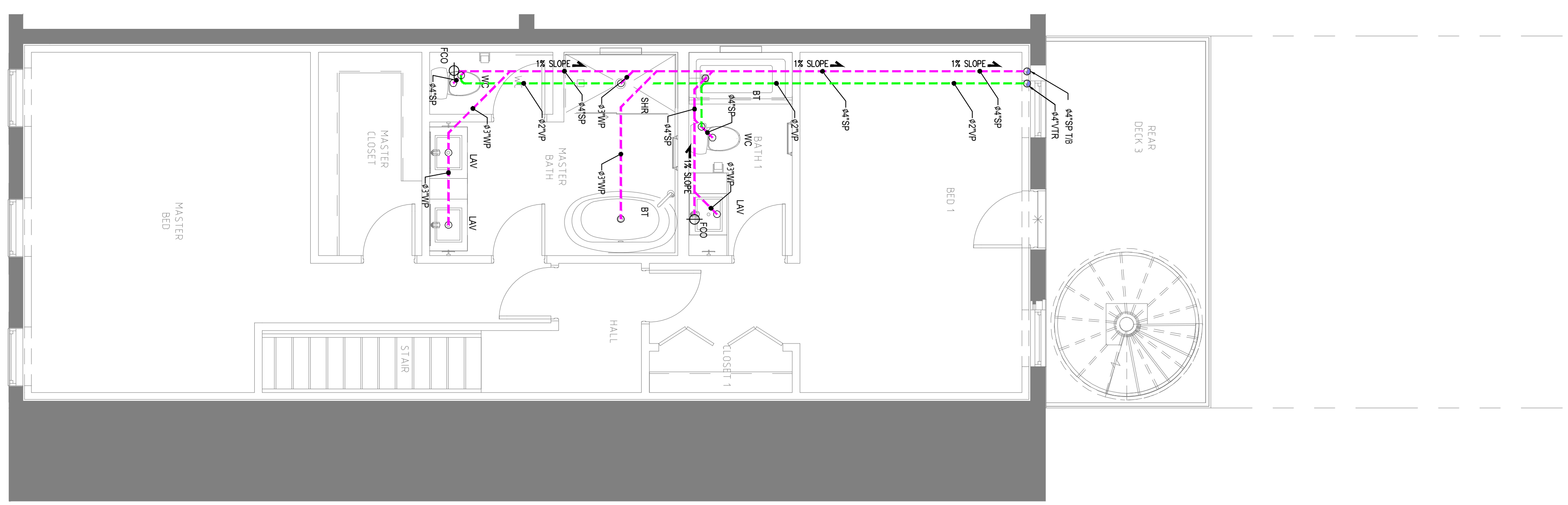
A3



PL 03 SECOND FLOOR DRAINAGE PLAN

SCALE 1/8"=1'-0"



PL 03 THIRD FLOOR DRAINAGE PLAN

SCALE 1/8"=1'-0"

PLUMBING LEGEND:	
SYMBOL	DESCRIPTION
	HWS - HOT WATER SUPPLY
	CWS - COLD WATER SUPPLY
	VP-VENT PIPE
	SP-SOIL PIPE
	WP-WASTE PIPE
	WP-WASTE PIPE FOR SUMP PUMP
	SHUT OFF VALVE
	GATE VALVE
WH	WATER HEATER
GT	GREASE TRAP
WC	WATER CLOSET
KS	KITCHEN SINK
AS	ABLUTION SPRAY
FCO	FLOOR CLEANOUT
FD	FLOOR DRAIN
LAV	LAVATORY

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DISAPPROVED	(D)

OWNER:

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DRAWING STATUS:

DISCIPLINE: PLUMBING

PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: SECOND & THIRD FLOOR DRAINAGE PLANS

DRAWN BY:	CHECKED BY:	DATE:
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JOB NO: 632-021	SHT NO: PL-03	SCALE: 1/8"=1'-0" @A3

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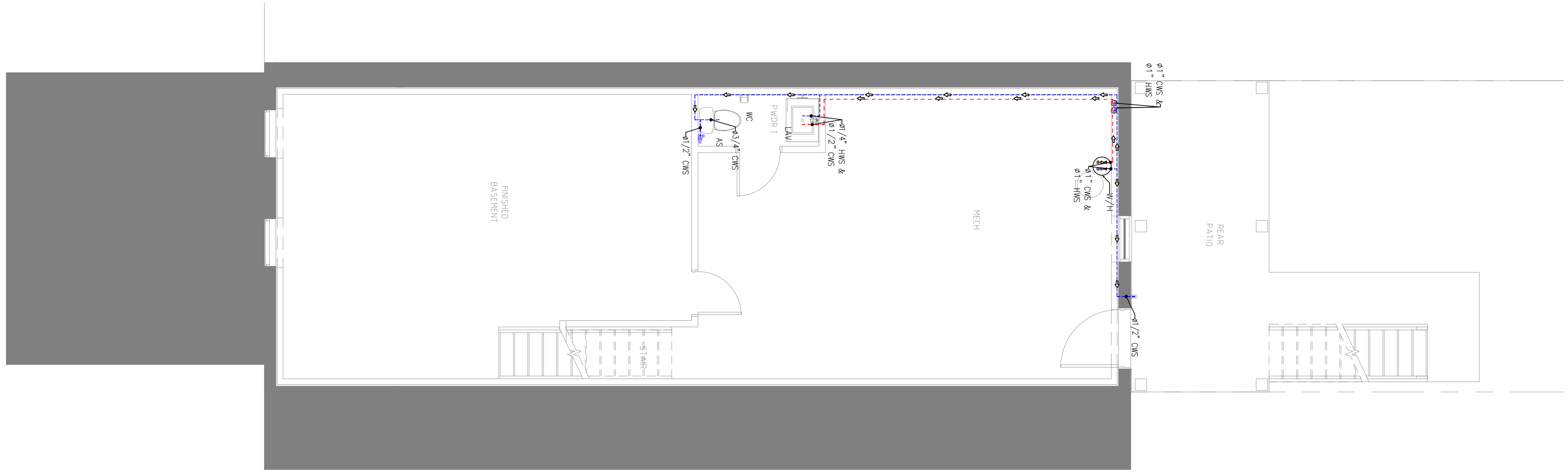
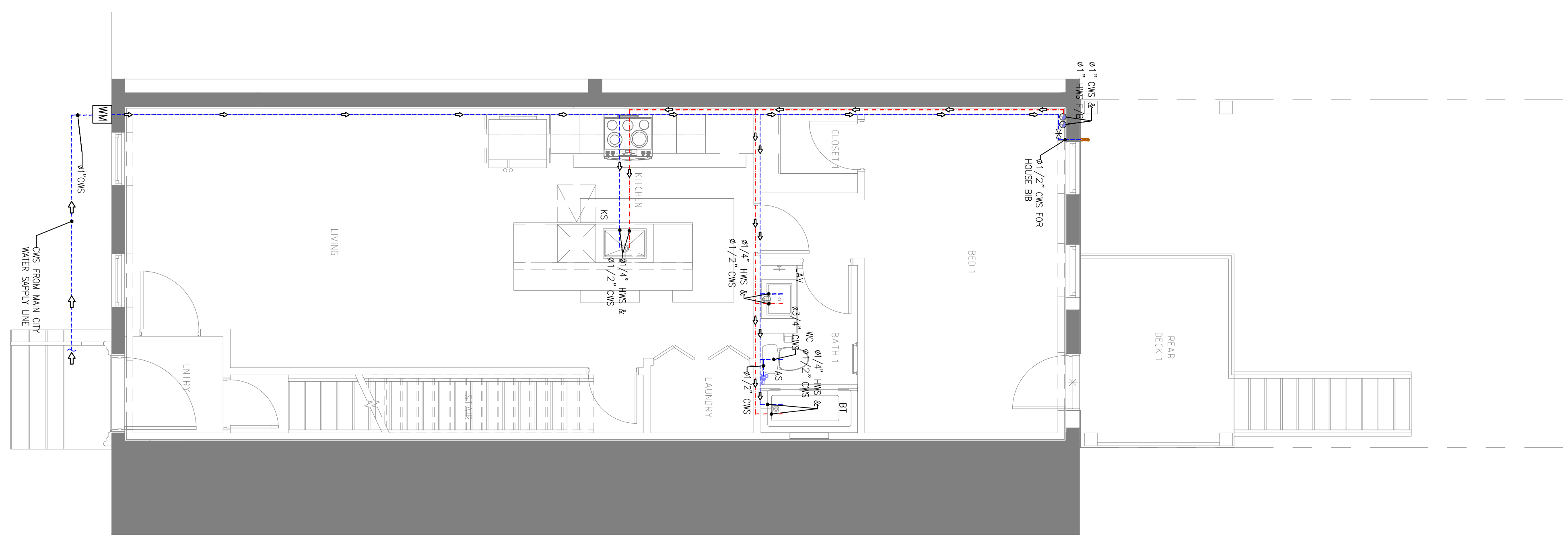


PL 04 **BASEMENT FLOOR WATER SUPPLY PLAN**
SCALE 1/8"=1'-0"



PL 04 **FIRST FLOOR WATER SUPPLY PLAN**
SCALE 1/8"=1'-0"

PLUMBING LEGEND:

SYMBOL	DESCRIPTION
(red dashed line)	HWS – HOT WATER SUPPLY
(blue dashed line)	CWS – COLD WATER SUPPLY
(green line)	VP–VENT PIPE
(magenta dashed line)	SP–SOIL PIPE
(magenta dashed line)	WP–WASTE PIPE
(orange dashed line)	WP–WASTE PIPE FOR SUMP PUMP
(valve symbol)	SHUT OFF VALVE
(valve symbol)	GATE VALVE
WH	WATER HEATER
GT	GREASE TRAP
WC	WATER CLOSET
KS	KITCHEN SINK
AS	ABLUTION SPRAY
FCO	FLOOR CLEANOUT
FD	FLOOR DRAIN
LAV	LAVATORY

OWNER SIGNATURE

REVISION STATUS

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APPROVED	**(A)**
APPROVED WITH NOTES	**(B)**
RESUBMIT	**(C)**
DISAPPROVED	**(D)**

OWNER:

ADDRESS:

DESIGN CONSULTANT:
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DRAWING STATUS:

DISCIPLINE: PLUMBING

PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: BASEMENT & FIRST FLOOR WATER SUPPLY PLANS

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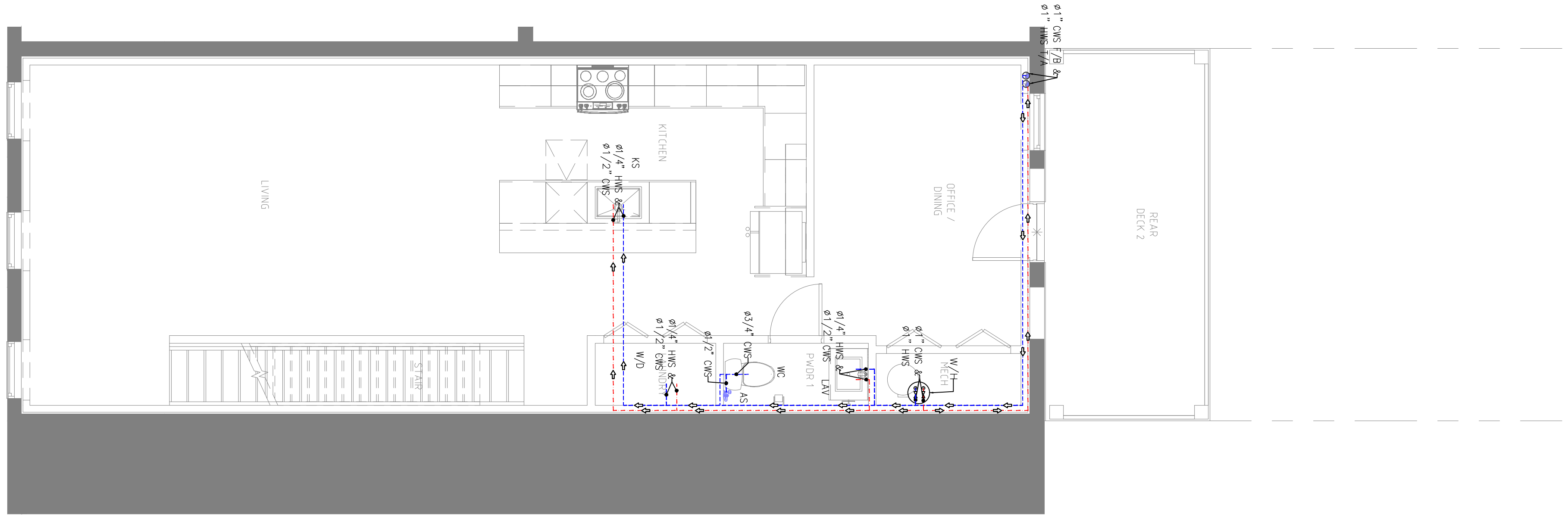
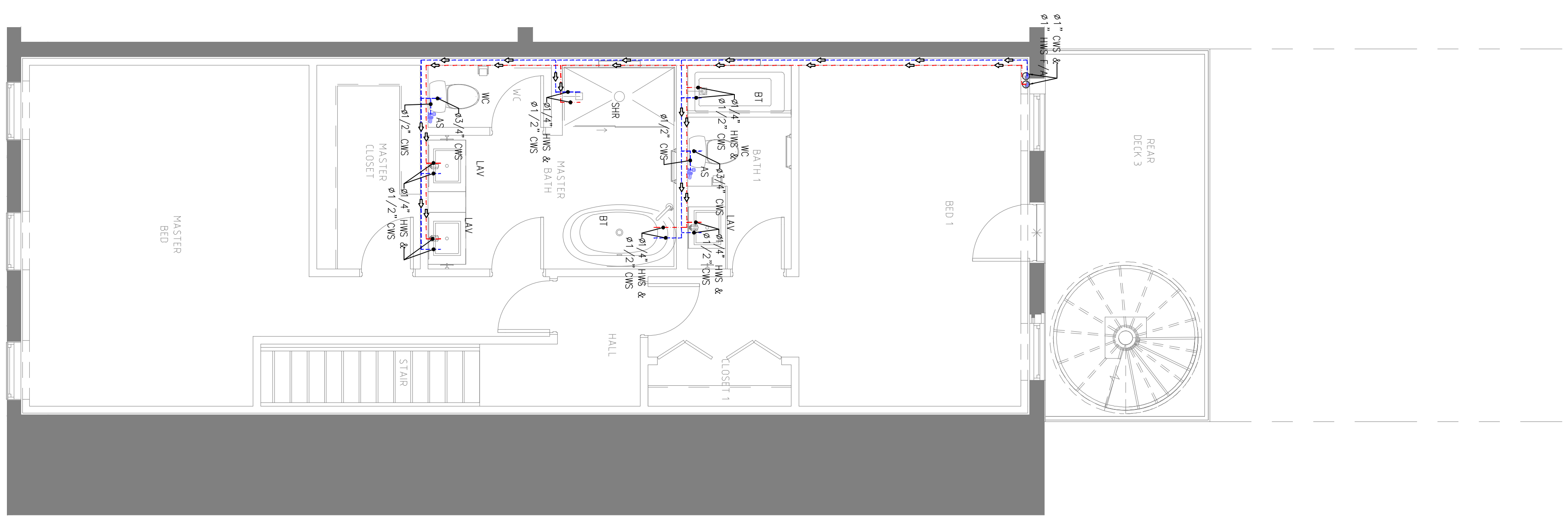


PL 05 **SECOND FLOOR WATER SUPPLY PLAN**
SCALE 1/8"=1'-0"



PL 05 **THIRD FLOOR WATER SUPPLY PLAN**
SCALE 1/8"=1'-0"

PLUMBING LEGEND:

SYMBOL	DESCRIPTION
	HWS – HOT WATER SUPPLY
	CWS – COLD WATER SUPPLY
	VP–VENT PIPE
	SP–SOIL PIPE
	WP–WASTE PIPE
	WP–WASTE PIPE FOR SUMP PUMP
	SHUT OFF VALVE
	GATE VALVE
WH	WATER HEATER
GT	GREASE TRAP
WC	WATER CLOSET
KS	KITCHEN SINK
AS	ABLUTION SPRAY
FCO	FLOOR CLEANOUT
FD	FLOOR DRAIN
LAV	LAVATORY

OWNER SIGNATURE

REVISION STATUS

Rev	DATE	DESCRIPTION	DRAWN BY
00	12-10-2023	ISSUED FOR APPROVAL	A.S
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CLIENT/CONSULTANT APPROVAL STATUS

APPROVED	(A)
APPROVED WITH NOTES	(B)
RESUBMIT	(C)
DISAPPROVED	(D)

OWNER:

ADDRESS:

DESIGN CONSULTANT:
TSABA CORP
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DRAWING STATUS:

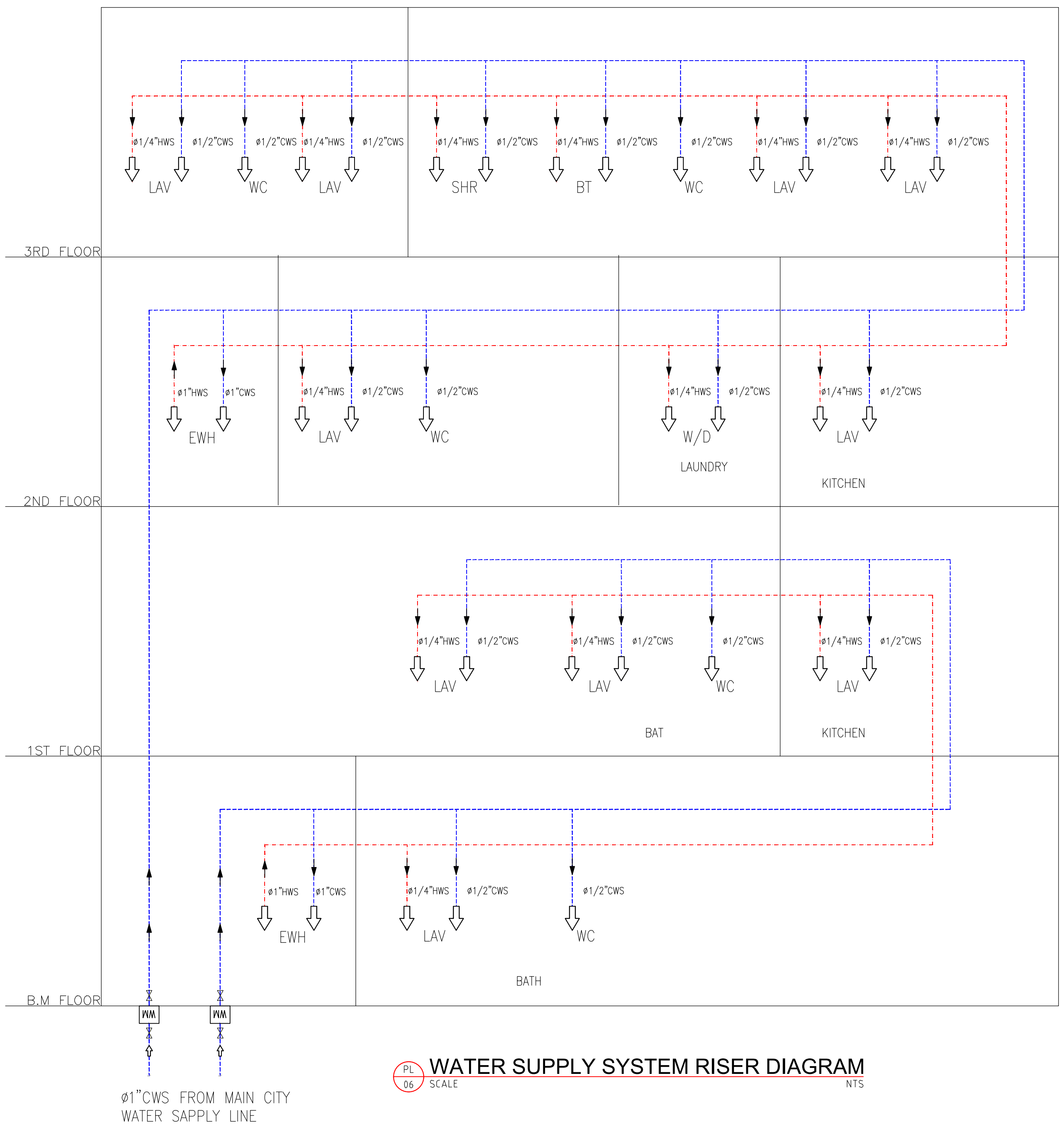
DISCIPLINE:
PLUMBING

PROJECT TITLE:
23-5550F_NCARROLLTON636

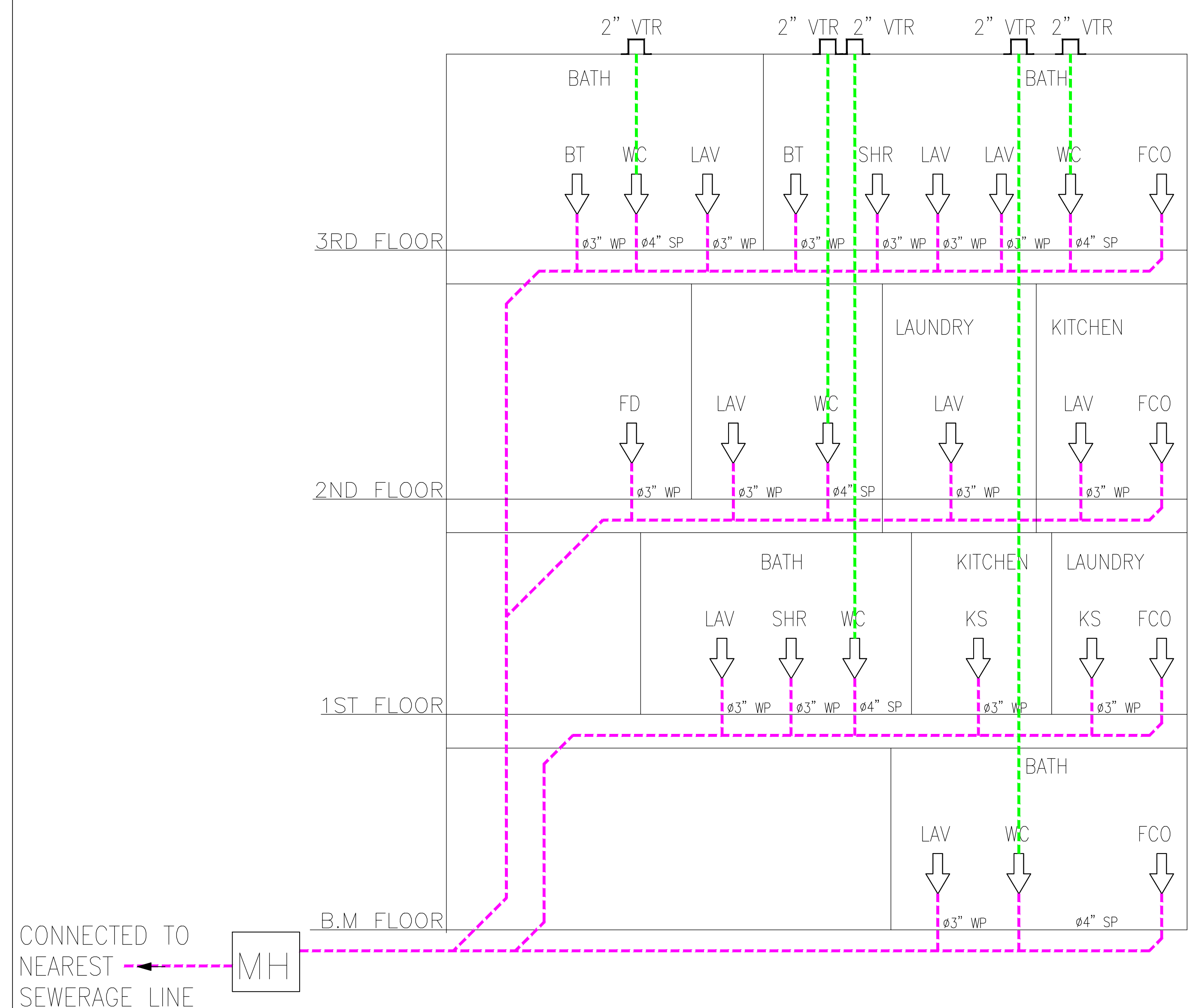
DRAWING TITLE:
SECOND & THIRD FLOOR WATER SUPPLY PLANS

DRAWN BY:	CHECKED BY:	DATE:
A.S	N.B	12 OCT. 2023
JOB NO 632-021	SHT NO PL-05	SCALE: 1/8"=1'-0" @A3

A3



PL 06 WATER SUPPLY SYSTEM RISER DIAGRAM
SCALE NTS



PL 06 DRAINAGE SYSTEM RISER DIAGRAM
SCALE NTS

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Rev	DATE	DESCRIPTION	DRAWN BY
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EMAIL:TS9DESIGNS@GMAIL.COM

DRAWING STATUS:

DISCIPLINE: PLUMBING

PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: RISER DIAGRAM

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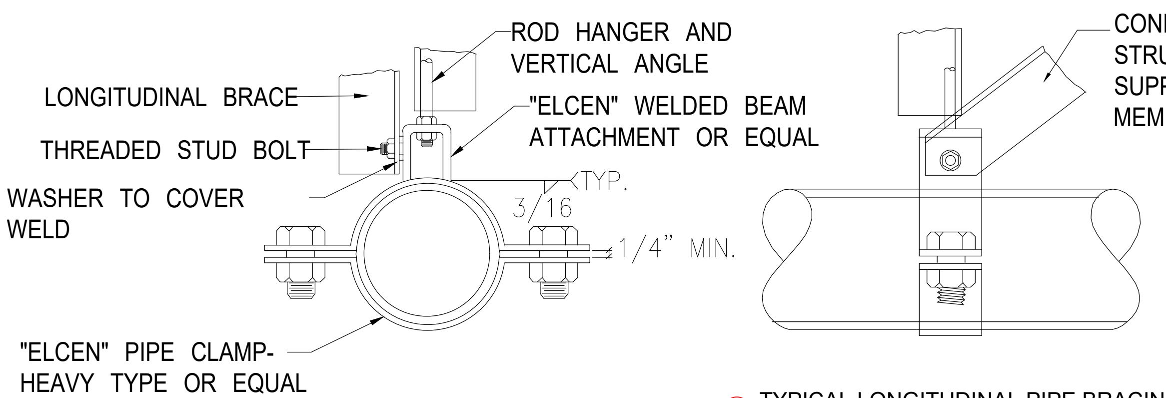
A.S	N.B	12 OCT. 2023
JOB NO 632-021	SHT NO PL-06	SCALE: NTS @A3

A3



D 01 TYPICAL LONGITUDINAL PIPE BRACING SCALE NTS

1.THIS BRACING DETAIL APPLIES ONLY FOR COLD WATER PIPE AND GAS PIPE WHERE MOVEMENT OF THE PIPE DUE TO TEMPERATURE DIFFERENTIAL IS NEGLIGIBLE

2.IT IS THE RESPONSIBILITY OF THE USER OF THIS GUIDELINE TO ASCERTAIN THAT AN ADEQUATE BRACING AND ANCHORAGE DEVICE BE DESIGNED FOR PIPE WHENEVER THE MOVEMENT DUE TO THERMAL DIFFERENTIAL EXISTS. PROVIDE ONE ANCHOR POINT IN ONE PIPE RUN. ALLOW FOR LONGITUDINAL PIPE MOVEMENT AT THE OPPOSITE END OF THE ANCHOR POINT.

D 01 LONGITUDINAL PIPE BRACING

SCALE NTS

GENERAL INFORMATION

This electric water heater's design is certified by Underwriters Laboratories (UL) and listed in accordance with UL 174. cUL listed in accordance with Canadian National Standard C22.2, No. 110.

Water heaters with a capacity of 19 gallons through 119 gallons are evaluated by UL in accordance with Part 280.707(d) (1) of HUD Mobile Home Construction and Safety Standards for Energy Efficiency. For mobile home installations, these water heaters must be installed in accordance with Section 3280.709 of HUD Mobile Home Construction and Safety Standards for Installation of Appliances.

This water heater must be installed in accordance with local codes. In the absence of local codes, install this water heater in accordance with the N.E.C. Reference Book (latest edition).

The warranty for this water heater is in effect only when the water heater is installed, adjusted, and operated in accordance with these Installation and Operating Instructions. The manufacturer will not be held liable for damage resulting from alteration and/or failure to comply with these instructions.

This water heater has been designed and certified for the purpose of heating potable water. The installation and use of this water heater for any purpose other than the heating of potable water, may cause damage to the water heater and create a hazardous condition and nullify the warranty.

This water heater can deliver scalding temperature water at any faucet in the system. Be careful whenever using hot water to avoid scalding injury. Certain appliances, such as dishwashers and automatic clothes washers, may require increased temperature water. By setting the thermostat on this water heater to obtain increased temperature water required by these appliances, you may create the potential for scald injury. To protect against injury, you should install an ASSE approved mixing valve in the water system. This valve will reduce point of discharge temperature by mixing cold and hot water in branch supply lines. Such valves are available from the manufacturer of this water heater or a local plumbing supplier. Please consult with a plumbing professional.

Water temperature over 125°F can cause severe burns instantly or death from scalds.

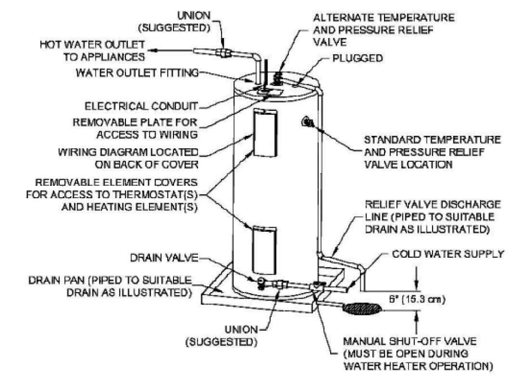
Children, disabled and elderly are at highest risk of being scalded.

Review this instruction manual before setting temperature at water heater.

Feel water before bathing or showering.

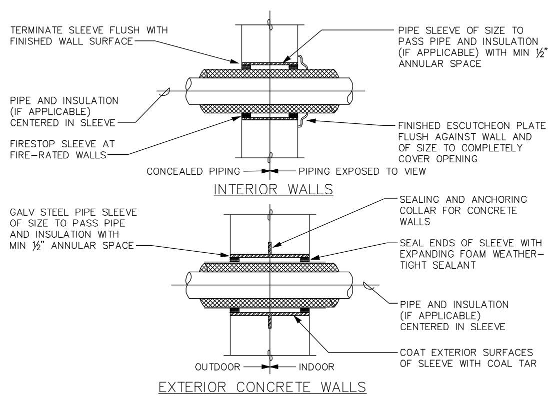
Temperature limiting valves are available.

APPROXIMATE TIME/TEMPERATURE RELATIONSHIPS IN SCALDS	
120°F (49°C)	More than 5 minutes
125°F (52°C)	1½ to 2 minutes
130°F (54°C)	About 30 seconds
135°F (57°C)	About 10 seconds
140°F (60°C)	Less than 5 seconds
145°F (63°C)	Less than 3 seconds
150°F (66°C)	About 1½ seconds
155°F (68°C)	About 1 second

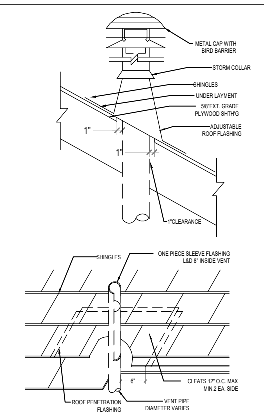


D 02 60 GALLON ELECTRIC WATER TANK SPECIFICATION

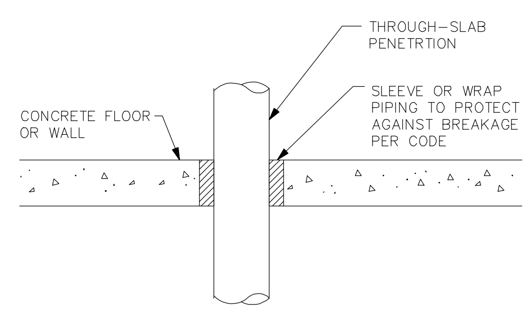
SCALE NTS



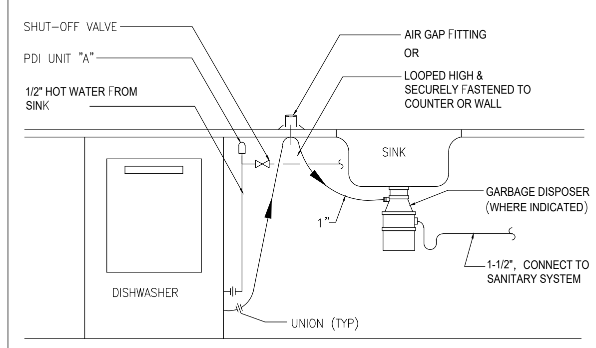
D 03 PIPE SLEEVES THROUGH WALLS SCALE NTS



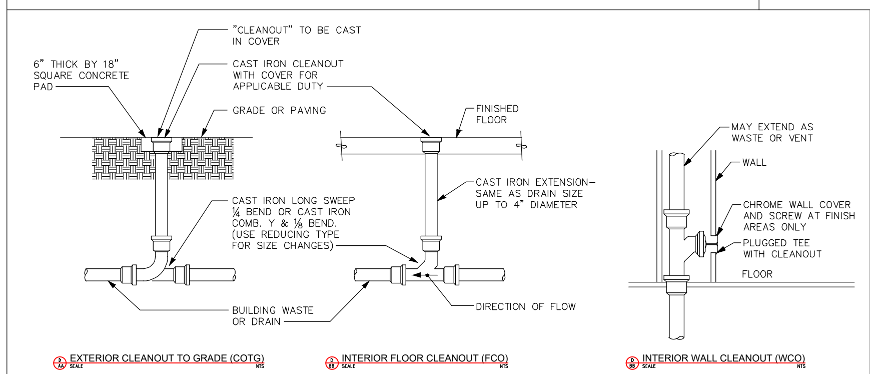
D 04 THRU ROOF VENT DETAIL SCALE NTS



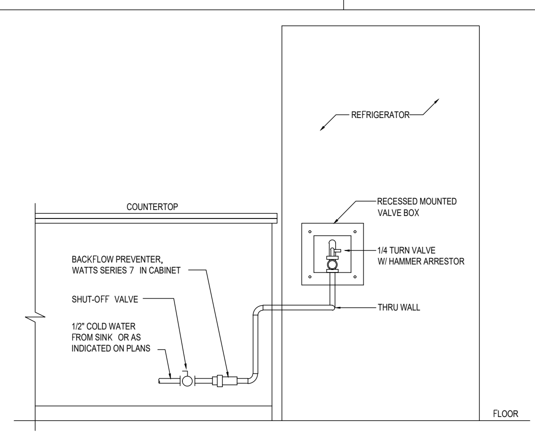
D 05 PIPE SLAB PENETRATION SCALE NTS



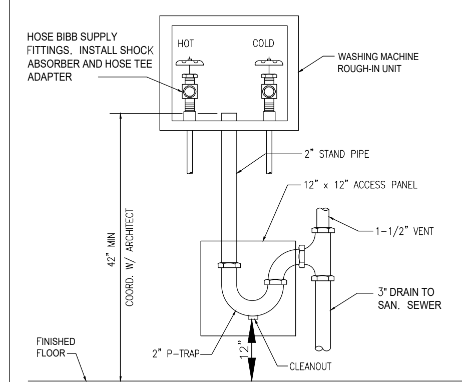
D 06 RESIDENTIAL DISHWASHER PIPING DETAIL SCALE NTS



D 07 CLEANOUT DETAILS SCALE NTS



D 08 REFRIGERATOR WATER CONNECTION DETAIL SCALE NTS



D 09 RESIDENTIAL WASHING MACHINE CONNECTION DETAIL SCALE NTS

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00	12-10-2023	ISSUED FOR APPROVAL	A.S
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CLIENT/CONSULTANT APPROVAL STATUS	
APPROVED	(A)
APPROVED WITH NOTES	(B)
RESUBMIT	(C)
DISAPPROVED	(D)

OWNER:

ADDRESS:

DESIGN CONSULTANT:
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DRAWING STATUS:

DISCIPLINE: PLUMBING

PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: DETAILS

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A.S	N.B	12 OCT. 2023
JOB NO 632-021	SHT NO PL-07	SCALE: NTS @A3

A3

MECHANICAL LEGEND

SYMBOL	ABBREV.	DESCRIPTION
E		COORDINATE WITH ELECTRICAL
CD	CD	CONDENSATE DRAIN (A.C)
D	D	DRAIN
RD	RD	REFRIGERANT DISCHARGE
RL	RL	REFRIGERANT LIQUID
RS	RS	REFRIGERANT SUCTION
		PIPE DOWN
		PIPE UP
UP		PIPE RISE (OR DN. FOR DROP)
		DIRECTION OF FLOW IN PIPE
	AV	AIR VENT (VALVE)
	CHV	CHECK VALVE
	CV (2W)	CONTROL VALVE (2-WAY)
	CV (3W)	CONTROL VALVE (3-WAY)
	FCD	AUTOMATIC FLOW CONTROL DEVICE
	SOV	SHUT OFF VALVE
		GLOBE/BALL/BUTTERFLY VALVE
	BV	COMBINATION BALANCING & SHUT-OFF VALVE
	FEV	FLOW ELEMENT VENTURI
		VALVE ON RISE OR DROP
	STR.	STRAINER
	CL	CAPPED LINE
	DN.	DOWN OR DROP
	UP	RISE OR RISER
	RV	PRESSURE RELIEF VALVE
	PG	PRESSURE GAUGE WITH BALL VALVE
	R.	ECCENTRIC REDUCER
	R.	CONCENTRIC REDUCER
	FC	FLEXIBLE CONNECTION (PIPE)
	PA	PIPE ANCHOR
	U	UNION
10x6		DUCTWORK (1ST NUMBER INDICATES WIDTH SHOWN), NET INSIDE DIMENSION
	TV	SQUARE ELBOW WITH TURNING VANES
		RADIUS ELBOW
	MVD	MANUAL VOLUME DAMPER
	MOD	MOTOR OPERATED DAMPER
	BDD	BACKDRAFT DAMPER
FD	FD	FIRE DAMPER
SD	SD	DUCT MOUNTED SMOKE DETECTOR
SFD	SFD	AUTOMATIC SMOKE AND FIRE DAMPER
	FLEX	FLEXIBLE CONNECTION (DUCTWORK)
	FLEX	FLEXIBLE CONNECTION OR SEISMIC JOINT

SYMBOL	ABBREV.	DESCRIPTION
		LINED DUCTWORK (OR PLENUM)
UP		DUCT RISE IN DIRECTION OF FLOW
DN		DUCT DROP IN DIRECTION OF FLOW
		ROUND DUCT UP
		ROUND DUCT DOWN
		SUPPLY DUCT UP
		SUPPLY DUCT DOWN
	RA/OA	RETURN AIR DUCT/OUTSIDE AIR DUCT UP
		RETURN AIR DUCT/OUTSIDE AIR DUCT DOWN
		EXHAUST AIR DUCT UP
		EXHAUST AIR DUCT DOWN
		DUCT TRANSITION
	CD	CEILING DIFFUSER
	RR	RETURN REGISTER
	ER	EXHAUST REGISTER
T AC-3	T'STAT	THERMOSTAT OR TEMPERATURE SENSOR (NUMBER INDICATES EQUIPMENTOR ZONE SERVED)
H	H'STAT	HUMIDISTAT
C2		CARBON DIOXIDE SENSOR
	CFM	CUBIC FEET PER MINUTE
OR		SYMBOL, SEE EQUIPMENT SCHEDULE
		SUPPLY AIR REGISTER
		RETURN AIR REGISTER
		SUPPLY AIR DIFFUSER W/ AIR DIRECTION
		SUPPLY AIR SIDEWALL DIFFUSER
		4-WAY SUPPLY AIR DIFFUSER
		RETURN/ EXHAUST AIR E GRILLE
		SUPPLY/EXHAUST AIR FAN
10x8 175 CFM D 2x		TERMINAL TAG/ QTY, AIR QUANTITY (CFM)
L		DOOR LOUVER
UC		UNDER CUT
		INTAKE/ EGRESS
		DUCT HEATER
		WASHROOM VENTILATOR

HVAC ABBREVIATIONS

A	AMPERES
AC	AIR CONDITIONING
AD	ACCESS DOOR
AFF	ABOVE FINISHED FLOOR
AL	ACOUSTICAL LINING
BHP	BRAKE HORSEPOWER
BTU	BRITISH THERMAL UNIT
BTUH	BTU PER HOUR
CD	CEILING DIFFUSER
CFM	CUBIC FEET PER MINUTE
CG	CEILING GRILLE
CLG	CEILING
COMPR	COMPRESSOR
CR	CEILING REGISTER
DB	DRY BULB
DIAM	DIAMETER
DN	DOWN
DWG	DRAWING
DX	DIRECT EXPANSION
EAT	ENTERING AIR TEMPERATURE
EDB	ENTERING DRY BULB TEMPERATURE
EF	EXHAUST FAN
EWB	ENTERING WET BULB
EWT	ENTERING WATER TEMPERATURE
°F	DEGREES FAHRENHEIT
FC	FLEXIBLE CONNECTION
FD	FIRE DAMPER
FIN FL	FINISHED FLOOR
FLA	FULL LOAD AMPERES
FPM	FEET PER MINUTE
FT	FEET
HD	HEAD
HR	HOUR
MAU	MAKE UP AIR UNIT
HZ	FREQUENCY
IN	INCH OR INCHES
KW	KILOWATT
LG	LENGTH
LAT	LEAVING AIR TEMPERATURE
LBS	POUNDS
LDB	LEAVING DRY BULB TEMPERATURE
LIN FT	LINEAR FEET
LWB	LEAVING WET BULB TEMPERATURE
MAX	MAXIMUM
MBH	THOUSAND BTU PER HOUR
MHP	MOTOR HORSEPOWER
MIN	MINIMUM
NIC	NOT IN CONTRACT
NO.	NUMBER
NTS	NOT TO SCALE
RA	RETURN AIR
RM	ROOM
RPM	REVOLUTIONS PER MINUTE
SP	STATIC PRESSURE
SPEC	SPECIFICATION
TEMP	TEMPERATURE
TG	TOP GRILLE
TV	TURNING VANES
TYP	TYPICAL
W	WIDTH
W/	WITH
W/O	WITHOUT
WB	WET BULB
WMS	WIRE MESH SCREEN
SG	SUPPLY GRILLE
RG	RETURN GRILLE
SP	SMOKE PURGE

OWNER SIGNATURE

REVISION STATUS

Rev	DATE	DESCRIPTION	DRAWN BY
00	12-10-2023	ISSUED FOR APPROVAL	A.S
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CLIENT/CONSULTANT APPROVAL STATUS	
APPROVED	(A)
APPROVED WITH NOTES	(B)
RESUBMIT	(C)
DISAPPROVED	(D)

OWNER:

ADDRESS:

DESIGN CONSULTANT:
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254 CHAPMAN RD, STE 208 #13130
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DRAWING STATUS:

DISCIPLINE: MECHANICAL

PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: NOTES

DRAWN BY:	CHECKED BY:	DATE:
A.S	N.B	12 OCT. 2023
JOB NO 632-021	SHT NO ME-01	SCALE: NTS @A3

A3

Goodman 3 Ton Heat Pump System

Brand Goodman	3 Ton
Nominal Capacity	Single
Stages	15.5 SEER
SEER Rating	11.5 EER
EER Rating	30000 BTU
Nominal Cooling Capacity (BTU)	30000 BTU
Heating Capacity	33000 BTU
Cooling Capacity	Goodman GSZ160601
Heat Pump unit	Goodman AVPTC61D14
Air Handler unit	Goodman HKSC20DB
Heat Kit unitGoodman	204121159
AHRI	Multi-Position
Configuration	1200 CFM
Max Airflow	Heating and Cooling
Operation	

Heat Pump Specifications

Product Line	GSZ16
Refrigerant	R-410a
Operation	Heating and Cooling
Installation Area	Outdoor
Stages	Single
Capacity	3 Ton
Cooling	BTU 36000
Cooling Capacity	36000 BTU
SEER Rating	16

Heat Pump Electrical Data

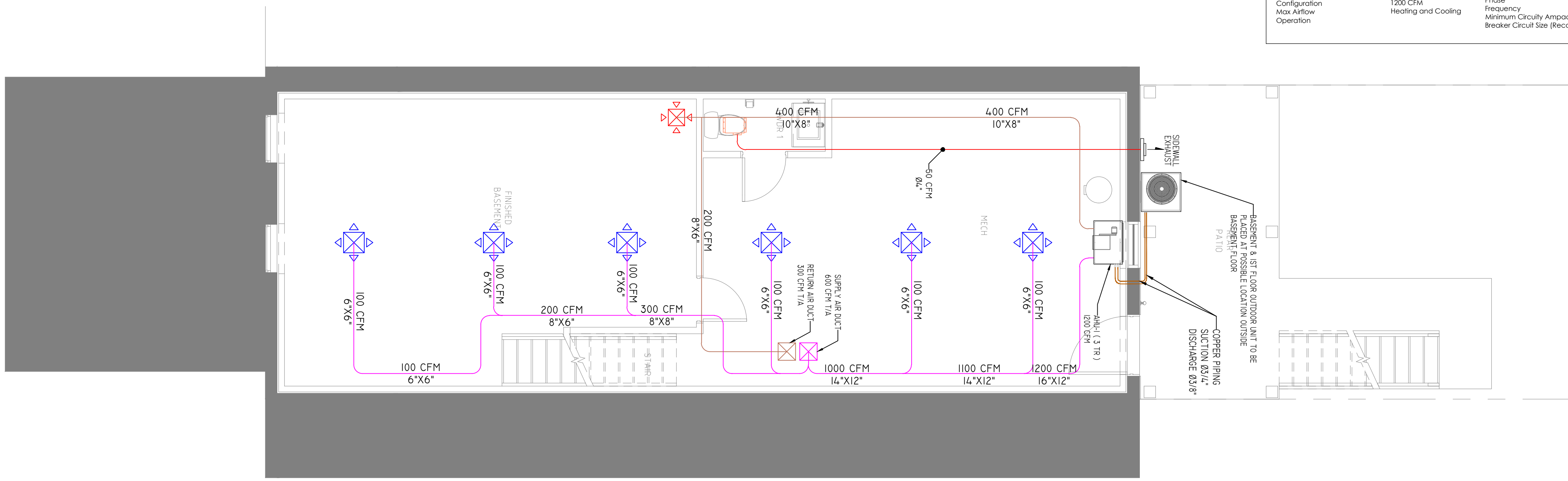
Voltage	208/230 Volts
Phase	1
Frequency	60 Hz
Minimum Circuity Ampacity	37 Amps
Breaker Circuit Size (Recommended)	60 Amps

Heat Pump Dimensions

Suction Conn. Size	1 1/8 Inches
Liquid Conn. Size	3/8 Inches
Heat Pump Height	40 Inches
Heat Pump Width	35.5 Inches
Heat Pump Depth	35.5 Inches

Heat Pump Weight

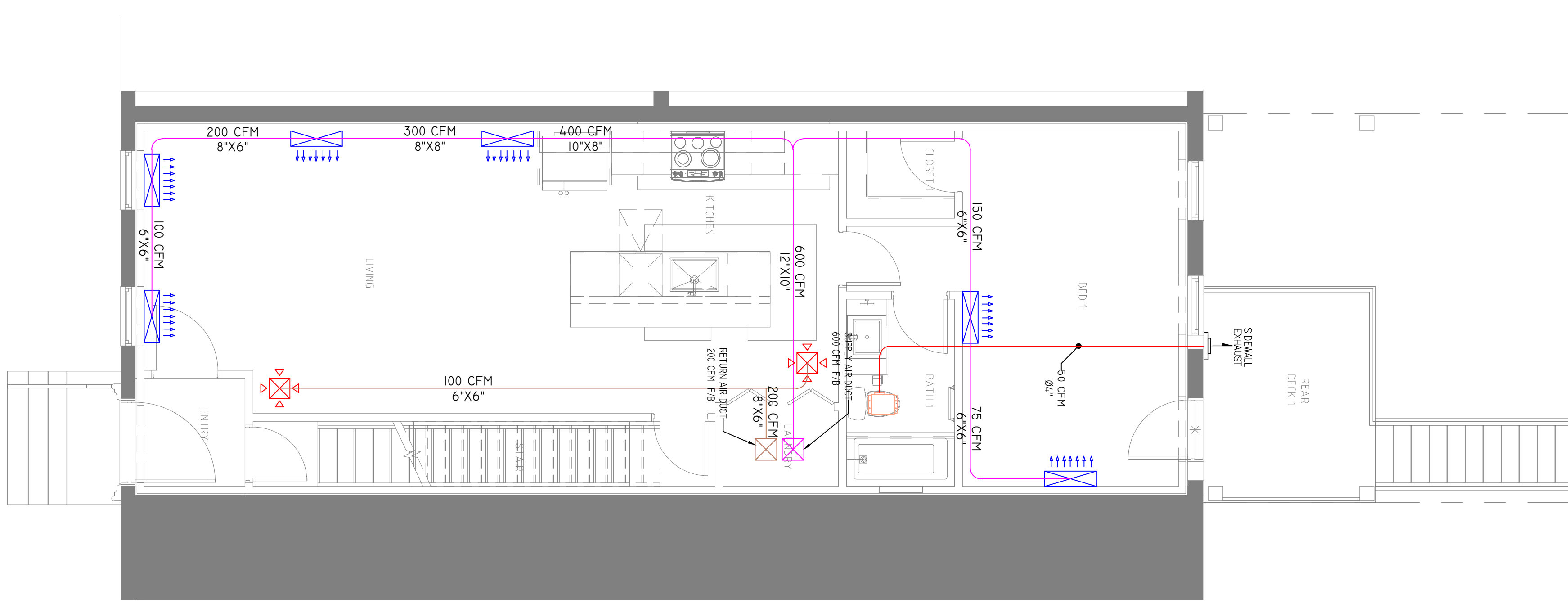
Product Weight	306 Pounds
Shipping Weight	326 Pounds



ME 02 BASEMENT FLOOR HVAC PLAN
SCALE 1/8"=1'-0"

SYMBOL	DESCRIPTION
	OUTDOOR AIR CONDENSING UNIT
	VERTICAL AIR HANDLER UNIT
	AIR SUPPLY DUCT. SIZE IS SHOWN ON PLAN
	AIR RETURN DUCT. SIZE IS SHOWN ON PLAN
	EXHAUST AIR DUCT. SIZE IS SHOWN ON PLAN
	COPPER PIPE. SIZE IS SHOWN ON PLAN
	SUPPLY AIR DIFFUSER
	SUPPLY AIR GRILL / VENT
	RETURN AIR DIFFUSER
	EXHAUST FAN
	WALL EXHAUST / GRILL

DRAWING LEGEND



ME 02 FIRST FLOOR HVAC PLAN
SCALE 1/8"=1'-0"

OWNER SIGNATURE

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APPROVED WITH NOTES	**(B)**
RESUBMIT	**(C)**
DISAPPROVED	**(D)**

OWNER:

ADDRESS:

DESIGN CONSULTANT:
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NEWARK, DELAWARE 19702
EMAIL:TS9DESIGNS@GMAIL.COM

DRAWING STATUS:

DISCIPLINE: MECHANICAL

PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: BASEMENT & FIRST FLOOR HVAC PLANS

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JOB NO 632-021	SHT NO ME-02	SCALE: 1/8"=1'-0" @A3

A3

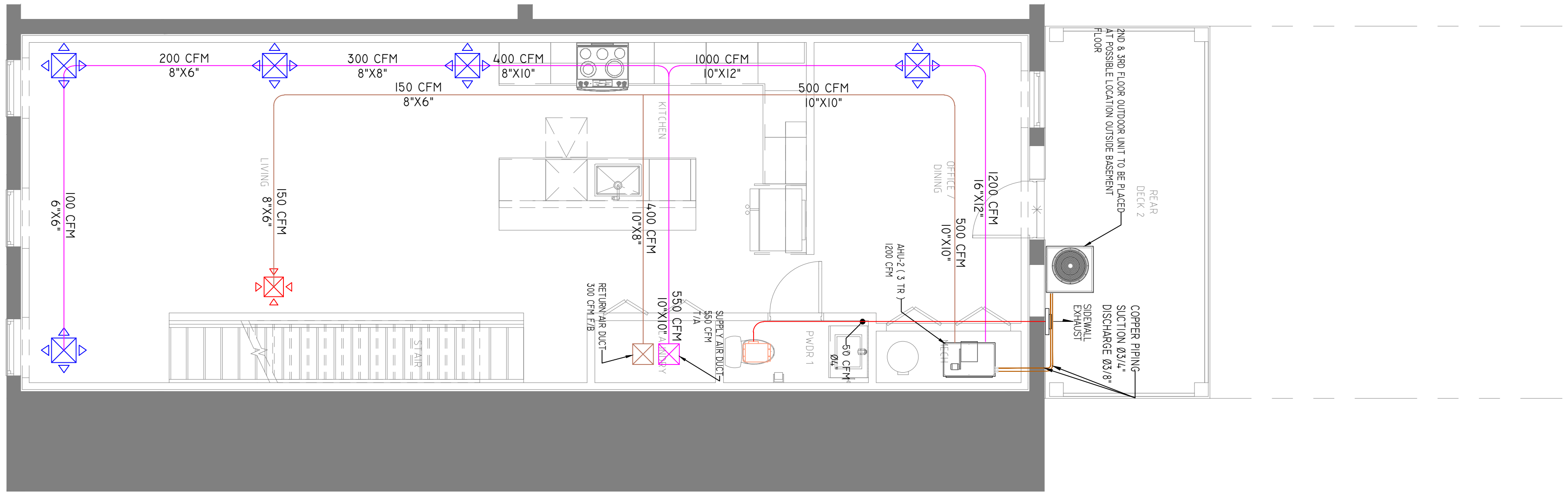
Goodman 3 Ton Heat Pump System	
Brand Goodman	3 Ton
Nominal Capacity	Single
Stages	15.5 SEER
SEER Rating	11.5 EER
EER Rating	30000 BTU
Nominal Cooling Capacity (BTU)	30000 BTU
Heating Capacity	33000 BTU
Cooling Capacity	Goodman GSZ160601
Heat Pump unit	Goodman AVPTC61D14
Air Handler unit	Goodman HKSC20DB
Heat Kit unitGoodman	204121159
AHRI	Multi-Position
Configuration	1200 CFM
Max Airflow	Heating and Cooling
Operation	

Heat Pump Specifications	
Product Line	GSZ16
Refrigerant	R-410a
Operation	Heating and Cooling
Installation Area	Outdoor
Stages	Single
Capacity	3 Ton
Cooling	BTU 36000
Cooling Capacity	36000 BTU
SEER Rating	16

Heat Pump Electrical Data	
Voltage	208/230 Volts
Phase	1
Frequency	60 Hz
Minimum Circuity Ampacity	37 Amps
Breaker Circuit Size (Recommended)	60 Amps

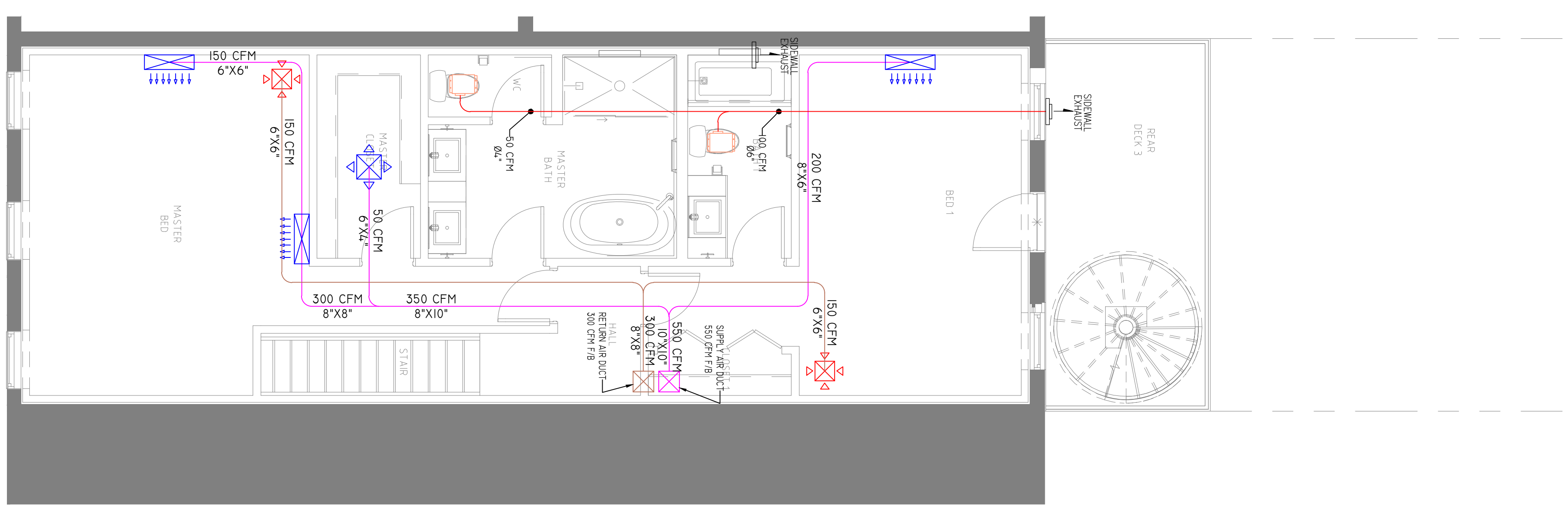
Heat Pump Dimensions	
Suction Conn. Size	1 1/8 Inches
Liquid Conn. Size	3/8 Inches
Heat Pump Height	40 Inches
Heat Pump Width	35.5 Inches
Heat Pump Depth	35.5 Inches

Heat Pump Weight	
Product Weight	306 Pounds
Shipping Weight	326 Pounds



ME 03 SECOND FLOOR HVAC PLAN
SCALE 1/8"=1'-0"

BASEMENT POSSIBLE



ME 03 THIRD FLOOR HVAC PLAN
SCALE 1/8"=1'-0"

DRAWING LEGEND

SYMBOL	DESCRIPTION
	OUTDOOR AIR CONDENSING UNIT
	VERTICAL AIR HANDLER UNIT
	AIR SUPPLY DUCT. SIZE IS SHOWN ON PLAN
	AIR RETURN DUCT. SIZE IS SHOWN ON PLAN
	EXHAUST AIR DUCT. SIZE IS SHOWN ON PLAN
	COPPER PIPE. SIZE IS SHOWN ON PLAN
	SUPPLY AIR DIFFUSER
	SUPPLY AIR GRILL / VENT
	RETURN AIR DIFFUSER
	EXHAUST FAN
	WALL EXHAUST / GRILL

OWNER SIGNATURE

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CLIENT/CONSULTANT APPROVAL STATUS

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RESUBMIT	**(C)**
DISAPPROVED	**(D)**

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254 CHAPMAN RD, STE 208 #13130
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EMAIL:TS9DESIGNS@GMAIL.COM

DRAWING STATUS:

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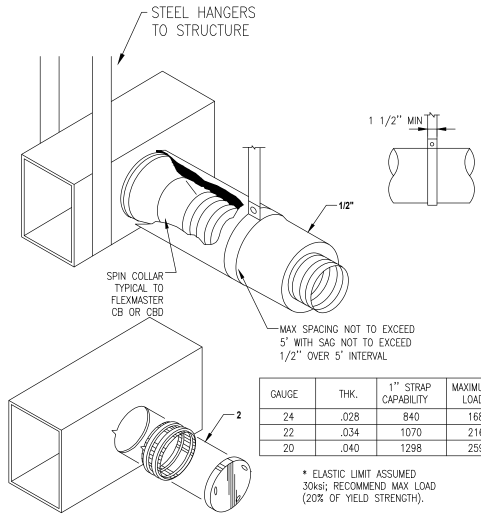
PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: SECOND & THIRD FLOOR HVAC PLANS

DRAWN BY:	CHECKED BY:	DATE:
A.S	N.B	12 OCT. 2023
JOB NO 632-021	SHT NO ME-03	SCALE: 1/8"=1'-0" @A3

A3



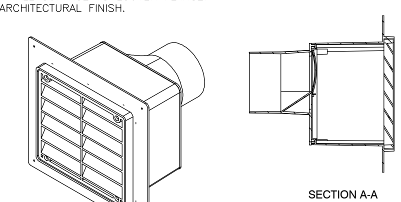
GAUGE	THK.	1" STRAP CAPABILITY	MAXIMUM LOAD
24	.028	840	168
22	.034	1070	216
20	.040	1298	259

* ELASTIC LIMIT ASSUMED 30ksi; RECOMMEND MAX LOAD (20% OF YIELD STRENGTH).

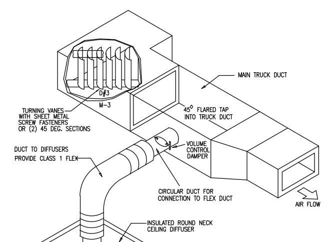
NOTES:
1. FLEX DUCT WITH EXTERNAL INSULATION AND VAPOR BARRIER JACKETING TYPICAL TO "FABRIFLEX"
2." INSPECTOR SERIES" SPIN DOORS
3. PROVIDE HANGER SPACING IAW FBCM AND SMASNA

D 01 SUPPORT REQUIREMENTS FOR DUCT
SCALE NTS

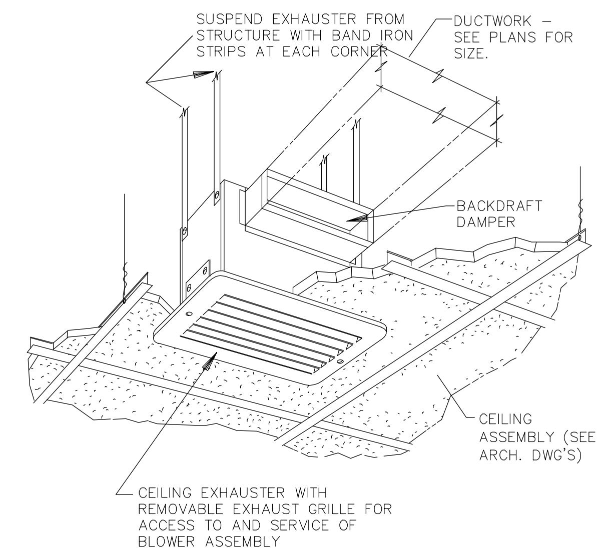
NOTES:
1. PROVIDE XVENT OR EQUAL NON-COORROSIVE WALL LOUVER W/ INTEGRAL NON-RETURN DAMPER RATED FOR MIAMI-DADE SERVICE.
2. MOUNT FLUSH WITH STRUCTURE AND SECURE W/ MASONARY ANCORS.
3. APPLY FINISHED WALL SURFACE TO LOUVER FLANGE.
4. PAINT TO SUIT ARCHITECTURAL FINISH.



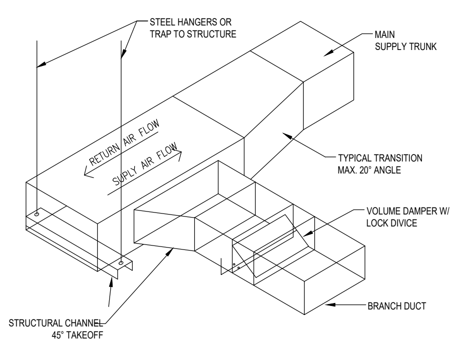
D 02 TYPICAL VENT LOUVER TERMINATION AT WEATHER
SCALE NTS



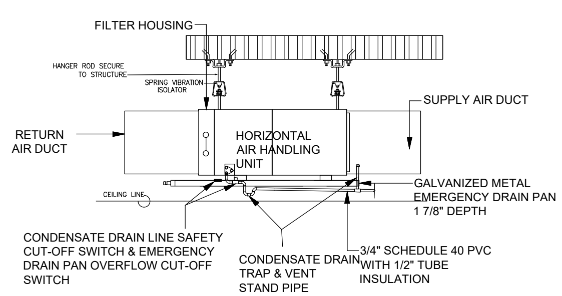
D 03 TURNING VANES DETAIL
SCALE NTS



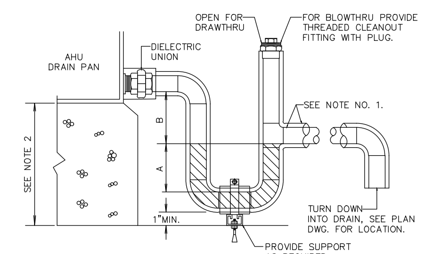
D 04 CEILING MOUNTED FAN DETAIL
SCALE NTS



D 05 BRANCH DUCT TAKEOFF DETAIL
SCALE NTS



D 06 AHU CEILING MOUNTED TYPICAL INSTALLATION DETAIL
SCALE NTS

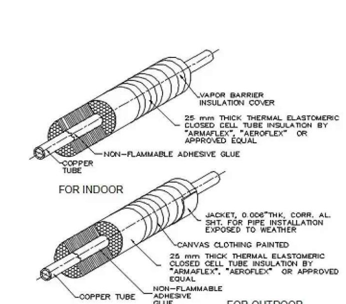


BLOWTHRU:
A = GREATER OF 4" OR 1/2" PLUS AHU TOTAL PRESSURE IN INCHES-WC.

DRAWTHRU:
B = GREATER OF 2" OR 1/2" PLUS AHU TOTAL PRESSURE IN INCHES-WC

NOTES:
1. REF. PLAN DWG'S FOR CONDENSATE DRAIN PIPE SIZE, (1"ø MIN. AT EACH AHU DRAIN CONNECTION POINT). REF. SPECIFICATIONS FOR PIPE AND INSULATION MATERIAL REQUIREMENTS.
2. PAD OR STANCHION HEIGHT DETERMINED BY SPACE REQD TO INSTALL TRAP, REF. PLAN DWG'S HEIGHT.

D 07 CONDENSATE DRAIN DETAIL
SCALE NTS

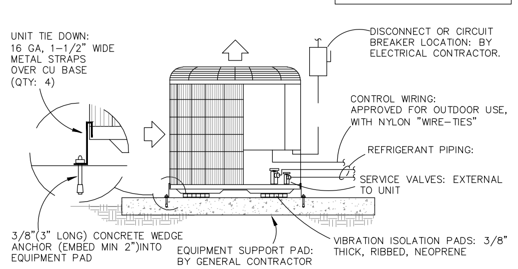


D 12 COPPER PIPING INSULATION DETAILS
SCALE

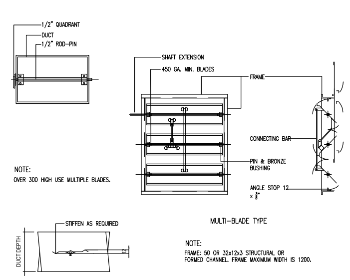
REFER TO THE STRUCTURAL DRAWINGS FOR MOUNTING AND ATTACHMENT DETAILS.

GROUND MOUTED CONDENSING UNIT SECURING REQUIRMENTS (FMC 301.12)
INSTALLATION SHALL WITHSTAND 170 MPH WINDLOAD AT EXPOSURE B
PROVIDE (1) #14 X 1" SCREW PER STRAP TO ATTACH STRAP TO CONDENSER

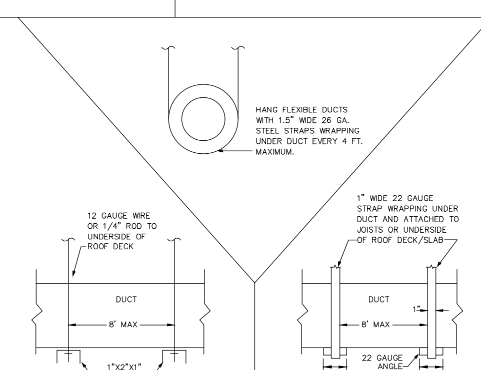
DETAIL COMPLIES WITH FBC MECH . 301.12



D 08 CONDENSER MOUNTING DETAIL
SCALE NTS



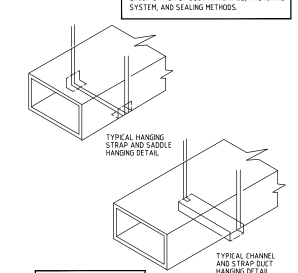
D 07 VOLUME DAMPER DETAIL
SCALE NTS



HANG ALL DUCT PER SMACNA STANDARDS

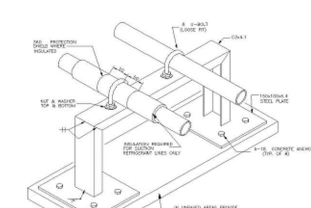
D 10 DUCT SUPPORT DETAIL
SCALE NTS

SEE DUCTWORK AND ACCESSORIES NOTES FOR SPECIFICATION OF DUCT MATERIALS, FASTENING SYSTEM, AND SEALING METHODS.



NOTE: ALL STRAP SHALL BE A MINIMUM OF 1-1/2,26 GA GALVINIZED STEEL. WITH A 6 FT MAXIMUM SPACING.

D 11 METHODS OF HANGING RIGID FIBERGLASS DUCT DETAIL
SCALE NTS



D 13 COPPER PIPING SUPPORT DETAILS
SCALE NTS

OWNER SIGNATURE

REVISION STATUS			
Rev	DATE	DESCRIPTION	DRAWN BY
00	12-10-2023	ISSUED FOR APPROVAL	A.S
01			
02			
03			
04			

CLIENT/CONSULTANT APPROVAL STATUS	
APPROVED	(A)
APPROVED WITH NOTES	(B)
RESUBMIT	(C)
DISAPPROVED	(D)

OWNER:

ADDRESS:

DESIGN CONSULTANT:
TSABA CORP
NADIF BRACEY
254 CHAPMAN RD, STE 208 #13130
NEWARK, DELAWARE 19702
EMAIL:TS9DESIGNS@GMAIL.COM

DRAWING STATUS:

DISCIPLINE: MECHANICAL

PROJECT TITLE: 23-5550F_NCARROLLTON636

DRAWING TITLE: DETAILS

DRAWN BY:	CHECKED BY:	DATE:
A.S	N.B	12 OCT. 2023
JOB NO 632-021	SHT NO ME-04	SCALE: NTS @A3